

PART I – PRACTICAL INFO

COMPUTATIONAL STATISTICAL MECHANICS



SAPIENZA
UNIVERSITÀ DI ROMA

ABOUT ME

Cristiano De Michele



RECEPTION HOURS:

Tuesday 11:00 – 12:00

(or by appointment)

Room F106 (first floor)

Dipartimento di Fisica

Nuovo Edificio «Fermi»

Please contact me in advance by email for an appointment.

Email: cristiano.demichela@uniroma1.it

Personal web page: <https://www.roma1.infn.it/~demichel/>

WHEN? – LECTURES AND LABORATORY

Lectures:

Monday	14:00-16:00	Aula Majorana
Friday	14:00-16:00	Aula Majorana

21/22 Lectures

WHEN? – LECTURES AND LABORATORY

Lectures:

Monday 14:00-16:00 Aula Majorana

Friday 14:00-16:00 Aula Majorana

Laboratory (practical classes):

Laboratorio di Calcolo «Fermi»

Tuesday 16:00-19:00

Starting from Tuesday 6 October

THERE WILL BE 10-11 **PRACTICAL CLASSES**

WHERE I WILL HELP YOU WRITE YOUR OWN SIMULATION CODE

POSSIBLY RELATED TO THE FINAL PROJECTS (**WITH YOUR LAPTOPS**

OR USING THE PC OF THE LABORATORY)

WHEN? – LECTURES AND LABORATORY

Lectures :

Monday 14:00-16:00 Aula Majorana

Friday 14:00-16:00 Aula Majorana

Laboratory (practical classes):

Laboratorio di Calcolo «Fermi»

Tuesday 16:00-19:00

**SIGN UP FOR THE COURSE ON ELEARNING SAPIENZA
PLATFORM (<https://elearning.uniroma1.it>)**

SEARCH FOR **CompStatMech**

DO THIS AS SOON AS YOU CAN!!!

WHEN? – LECTURES

Lectures :

Monday	14:00-16:00	Aula Majorana
Friday	14:00-16:00	Aula Majorana

Laboratory (practical classes):

Laboratorio di Calcolo «Fermi»

Tuesday 16:00-19:00

FILL THE QUESTIONNAIRE WHICH YOU CAN FIND ON THE
ELEARNING COURSE WEBPAGE

TOP OF THE PAGE CLICK ON

Computational Statistical Mechanics 2026-27 Questionnaire

DO THIS WITHIN THIS WEEK!!!

FINAL EXAMINATION

The final evaluation consists of an **oral exam on at least one topic** among the ones discussed during the course and which will be chosen by the student. The list of possible topics will be provided by the teacher before the exams. If you prepare **just one topic** your **final grade** of the oral exam will be **at most 25/30**. The students are also encouraged to work on a project involving the development of a simulation code and to prepare a term paper. The term paper may earn you a **bonus up to 4 points**. To receive honors (30 cum laude), you must achieve a grade greater than or equal to 32.

Final Examination Sessions (on infostud)

Winter 2027:	January 2027 February 2027
Spring 2027:	May 2027
Summer 2027:	June 2027 July 2027
Fall 2027:	September 2027 November 2027

COURSE GOALS

- Learn the **basic numerical techniques** of **Molecular Dynamics (MD)** and **Monte Carlo (MC)** to write a simulation code in C++ to investigate static and dynamic properties of a complex system.
- The student will be able to optimally **exploit common simulation packages** for investigating complex systems (including colloidal and biomolecular systems) since he/she will have acquired a deep understanding of algorithms and techniques on which these tools rely.

COURSE PROGRAMME

LECTURES

- Recalls of classical statistical mechanics
- MC Methods, Metropolis MC (NTV, NPT, ...), biased MC
- Typical interaction potentials
- Linked cell list, neighbour list, initial conditions
- MD, symplectic algorithms
- Multiple-time-step integrators
- Extended ensemble for controlling temperature and pressure
- Algorithms for handling holonomic constraints
- Brownian dynamics

LABORATORY

- Object-oriented paradigm: encapsulation, inheritance, overloading and generic programming
- Introduction to C++ programming language
- Implementing a simulation in C++
- Python: an effective tool to analyze data from computer simulations

REFERENCES – LECTURES

1. *Understanding Molecular Simulation*
D. Frenkel and B. Smit, Academic Press
2. *Statistical Mechanics: Theory and Molecular Simulation*
Mark Tuckerman Oxford Graduate Press - Oxford
3. *Computer Simulation of Liquids (new edition)*
M. P. Allen and D. J. Tildesley, Clarendon Press Oxford
4. *The Art of Molecular Dynamics Simulation*
D. C. Rapaport, Cambridge University Press
5. *Theory of Simple Liquids*
J.-P. Hansen and I. R. McDonald, Academic Press
6. Course notes on **google classroom** (see <https://elearning.uniroma1.it>)

REFERENCES – LABORATORY

1. **C++ How to program (5th Edition), H. Deitel and P. Deitel, Prentice Hall**
[*Introductory book, suitable for C++ beginners*]
2. **The C++ Programming Language (4th Editions), Bjarne Stroustrup, Addison-Wesley Professional**
[*This book is the equivalent of Kernighan-Ritchie for C programming language*]
3. **Effective Modern C++, Scott Meyers, O'Reilly Media**
[*Intermediate level*]

PART II – INTRODUCTION

OUTLINE

- BRIEF HISTORY OF COMPUTERS
- SIMULATION OF COMPLEX SYSTEMS
- MODELS

OUTLINE

- **BRIEF HISTORY OF COMPUTERS**
- **SIMULATION OF COMPLEX SYSTEMS**
- **MODELS**

WHY DO WE NEED COMPUTERS?

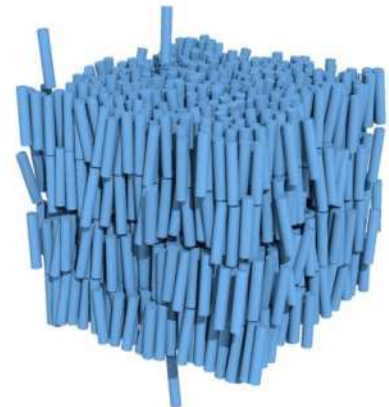
**TO BUILD THE HYDROGEN BOMB
(Nicholas Metropolis, MANIAC, 1951-52)**



TO PLAY (XBOX, PLAYSTATION,...)



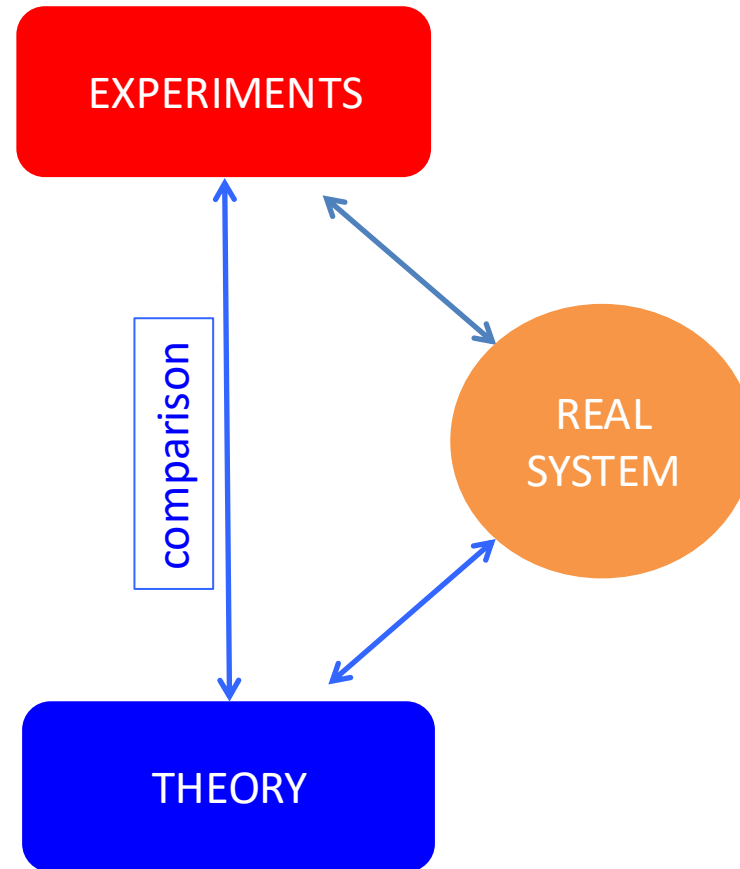
**TO CARRY OUT SIMULATIONS
OF COMPLEX SYSTEMS
(METROPOLIS ALGORITHM, 1953)**



THEORY AND EXPERIMENTS

I.E. WHAT HAPPENED BEFORE (OR WITHOUT) COMPUTERS

BEFORE 50s...



MECHANICAL COMPUTERS: WORKING FREQUENCY


$$\tau = \sqrt{\frac{m\sigma^2}{E}}$$

$$m = 0.1\text{kg}$$

$$\sigma = 0.01\text{ m}$$

$$E = 1\text{ J}$$

$$\tau = 3\text{ ms}$$

$$\nu \approx 300\text{ Hz}$$

THEORY AND EXPERIMENTS

I.E. WHAT HAPPENED BEFORE (OR WITHOUT) COMPUTERS

- Theories but only few exact and most of them approximate
- Bernal did mechanical (manual) simulations of large assemblies of macroscopic spheres (e.g. rubber balls and ball bearings)



J.D. Bernal

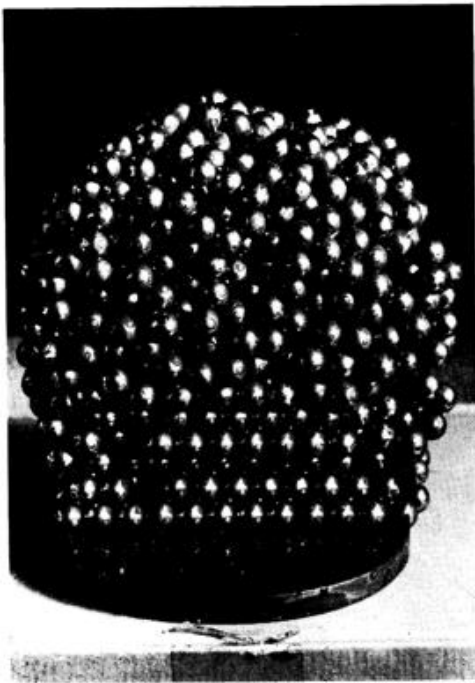


FIGURE 16. Ball-bearing assembly showing transition from random close-packing to regular crystalline array induced by inserting a flat plate.

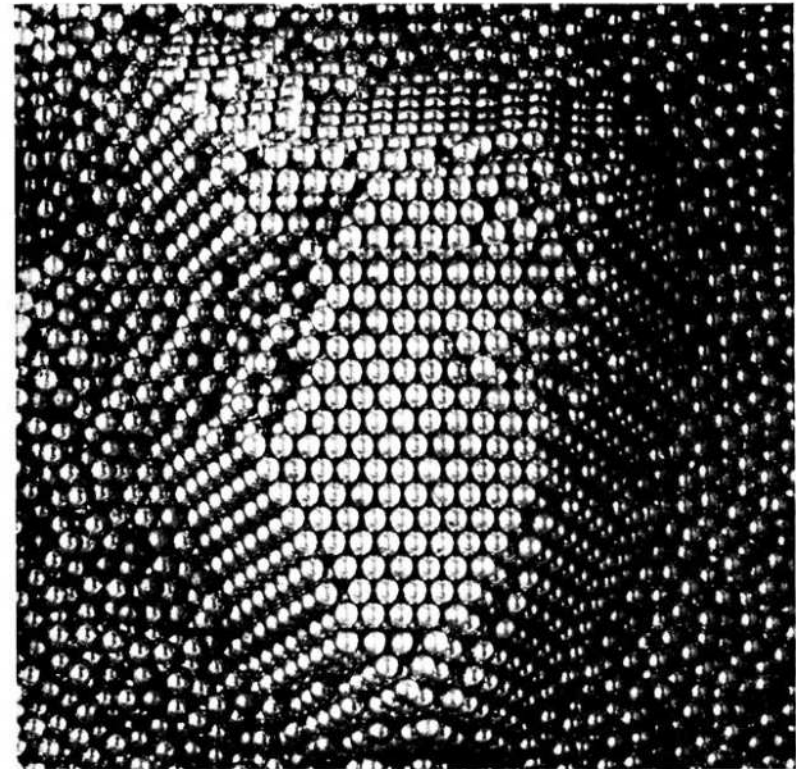
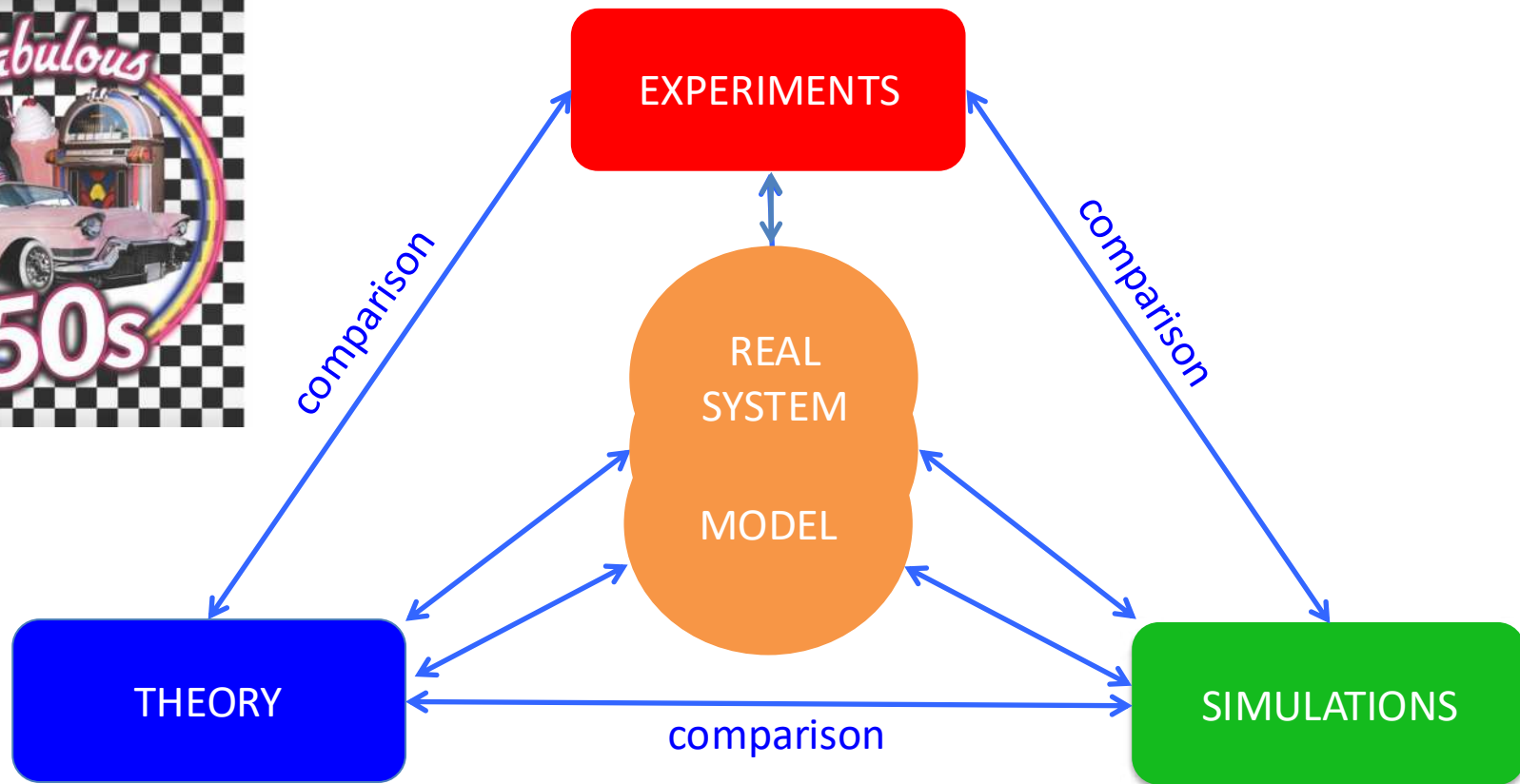


FIGURE 14. Face-centred cubic 'crystal' surrounded by 'liquid' caused by shearing ball-bearing mass. 111 face is shown at the top surface. 19

SIMULATIONS

DURING 50s A THIRD WAY EMERGED...



ELECTRONIC CALCULATORS: WORKING FREQUENCY



AMD Ryzen 9950
(4.3 Ghz / 5.7 Ghz)



Intel Core i9 14900KF
(2.4 Ghz / 6 Ghz)

SUPER-BRIEF HISTORY OF COMPUTER SIMULATIONS

- 1939-1945 (Second World War): **Nuclear Weapon Development** and **Code Breaking**
- 1950s: Electronic **computers** available for **non-military** use
- 1952: Los Alamos **MANIAC** became operational (under the lead of N. **Metropolis**)
- 1953: Milestone publication of **Metropolis** on **Monte Carlo (MC)** simulations

THE JOURNAL OF CHEMICAL PHYSICS

VOLUME 21, NUMBER 6

JUNE, 1953

Equation of State Calculations by Fast Computing Machines

NICHOLAS METROPOLIS, ARIANNA W. ROSENBLUTH, MARSHALL N. ROSENBLUTH, AND AUGUSTA H. TELLER,
Los Alamos Scientific Laboratory, Los Alamos, New Mexico

AND

EDWARD TELLER,* *Department of Physics, University of Chicago, Chicago, Illinois*

(Received March 6, 1953)

A general method, suitable for fast computing machines, for investigating such properties as equations of state for substances consisting of interacting individual molecules is described. The method consists of a modified Monte Carlo integration over configuration space. Results for the two-dimensional rigid-sphere system have been obtained on the Los Alamos MANIAC and are presented here. These results are compared to the free volume equation of state and to a four-term virial coefficient expansion.

SUPER-BRIEF HISTORY OF COMPUTER SIMULATIONS

- 1957: First **Molecular Dynamics** (MD) simulation of **Hard Spheres**
- 1957: Meeting in New Jersey (USA) with 15 scientists (2 Nobel laureates) – Can **hard spheres** form a **stable crystal**?
Half the audience answered **YES**
Phase Transition for a Hard Sphere System
- Work of Alder and Wainwright proved the predictive power of computers

B. J. ALDER AND T. E. WAINWRIGHT

University of California Radiation Laboratory, Livermore, California

(Received August 12, 1957)

A CALCULATION of molecular dynamic motion has been designed principally to study the relaxations accompanying various nonequilibrium phenomena. The method consists of solving exactly (to the number of significant figures carried) the simultaneous classical equations of motion of several hundred particles by means of fast electronic computers. Some of the details as they relate to hard spheres and to particles having square well potentials of attraction have been described.^{1,2} The method has been used also to calculate equilibrium properties, particularly the equation of state of hard spheres where differences with previous Monte Carlo³ results appeared.

The calculation treats a system of particles in a rectangular box with periodic boundary conditions.⁴

SUPER-BRIEF HISTORY OF COMPUTER SIMULATIONS

- 1959: First MD simulation of real material by Vineyard group (radiation damage on metallic copper)

PHYSICAL

15, 1960

The coordinates and velocities of all atoms at alternate time steps are stored on magnetic tape and can be printed out as desired. The positions of selected atoms can also be displayed as dots on a cathode-ray screen. Displays are presented sequentially and multi-flash pictures, such as Figs. 14 and 15, can be taken on stationary film; by advancing the film after each display, moving pictures have been made.

Localized vibrational modes associated with interstitials, agitations qualitatively like thermal spikes, ring annealing processes, and a higher energy process somewhat like a displacement spike have been observed. Replacements have been found to be very numerous.

The configurations of various static defects have also been studied in this model. The interstitial is found to reside in a "split" configuration, sharing a lattice site with another atom. The crowdion is found not to be stable, and Frenkel pairs are stable only beyond minimum separations, which are found to be very much dependent on orientation.

SUPER-BRIEF HISTORY OF COMPUTER SIMULATIONS

- 1964: First MD simulation of liquid Argon by Rahman

PHYSICAL REVIEW

VOLUME 136, NUMBER 2A

19 OCTOBER 1964

Correlations in the Motion of Atoms in Liquid Argon*

A. RAHMAN

Argonne National Laboratory, Argonne, Illinois

(Received 6 May 1964)

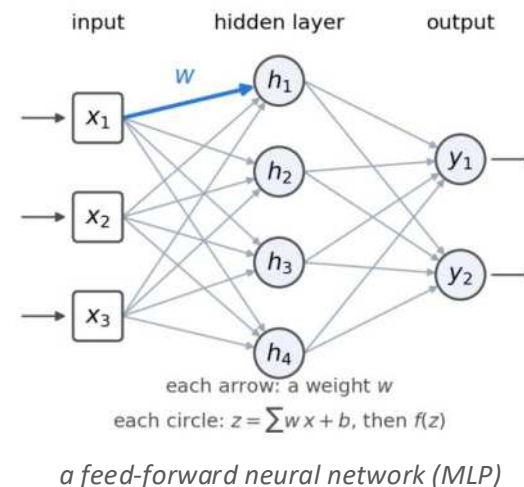
A system of 864 particles interacting with a Lennard-Jones potential and obeying classical equations of motion has been studied on a digital computer (CDC 3600) to simulate molecular dynamics in liquid argon at 94.4°K and a density of 1.374 g cm⁻³. The pair-correlation function and the constant of self-diffusion are found to agree well with experiment; the latter is 15% lower than the experimental value. The spectrum of the velocity autocorrelation function shows a broad maximum in the frequency region $\omega = 0.25(k_B T / \hbar)$. The shape of the Van Hove function $G_s(r, t)$ attains a maximum departure from a Gaussian at about $t = 3.0 \times 10^{-12}$ sec and becomes a Gaussian again at about 10^{-11} sec. The Van Hove function $G_d(r, t)$ has been compared with the convolution approximation of Vineyard, showing that this approximation gives a too rapid decay of $G_d(r, t)$ with time. A delayed-convolution approximation has been suggested which gives a better fit with $G_d(r, t)$; this delayed convolution makes $G_d(r, t)$ decay as t^4 at short times and as t at long times.

- After that widespread use of computers
- Nevertheless basic algorithms have been developed in the 1950s!

SUPER-BRIEF HISTORY OF COMPUTER SIMULATIONS

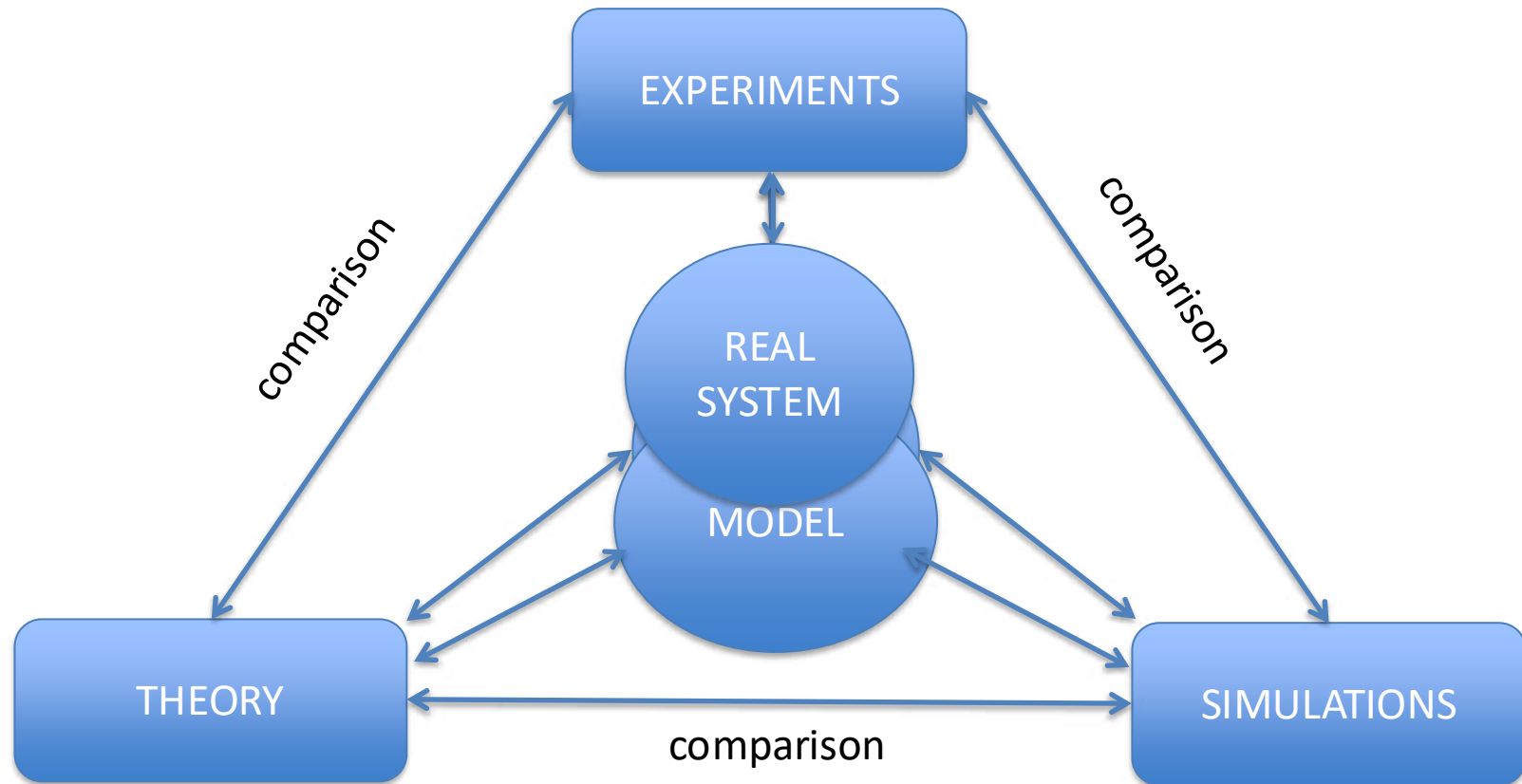
THE MACHINE-LEARNING ERA

- **1985 Car–Parrinello:** ab initio MD, forces from DFT “on the fly” PRL 55, 2471
- **2007 Behler–Parrinello:** first high-dimensional **neural-network potential** PRL 98, 146401
- **2018 DeePMD / DeePCG** (Car, E et al.): water with DFT accuracy; first deep-NN coarse-grained model PRL 120, 143001; JCP 149, 034101
- **2019–20 CGnet / CGSchNet** (Clementi, Noé et al.): ML coarse-grained proteins by force matching + priors ACS Cent. Sci. 5, 755; JCP 153, 194101; Nat. Chem. 17, 1284 (2025)
- **2020 Gordon Bell Prize** to DeePMD-kit: 10^8 atoms with ab initio accuracy
- **2024 Nobel Prizes:** Hopfield & Hinton (Physics); Baker, Hassabis & Jumper (Chemistry)



ML changes the potential, not the algorithms: MD and MC are still those of the 1950s!

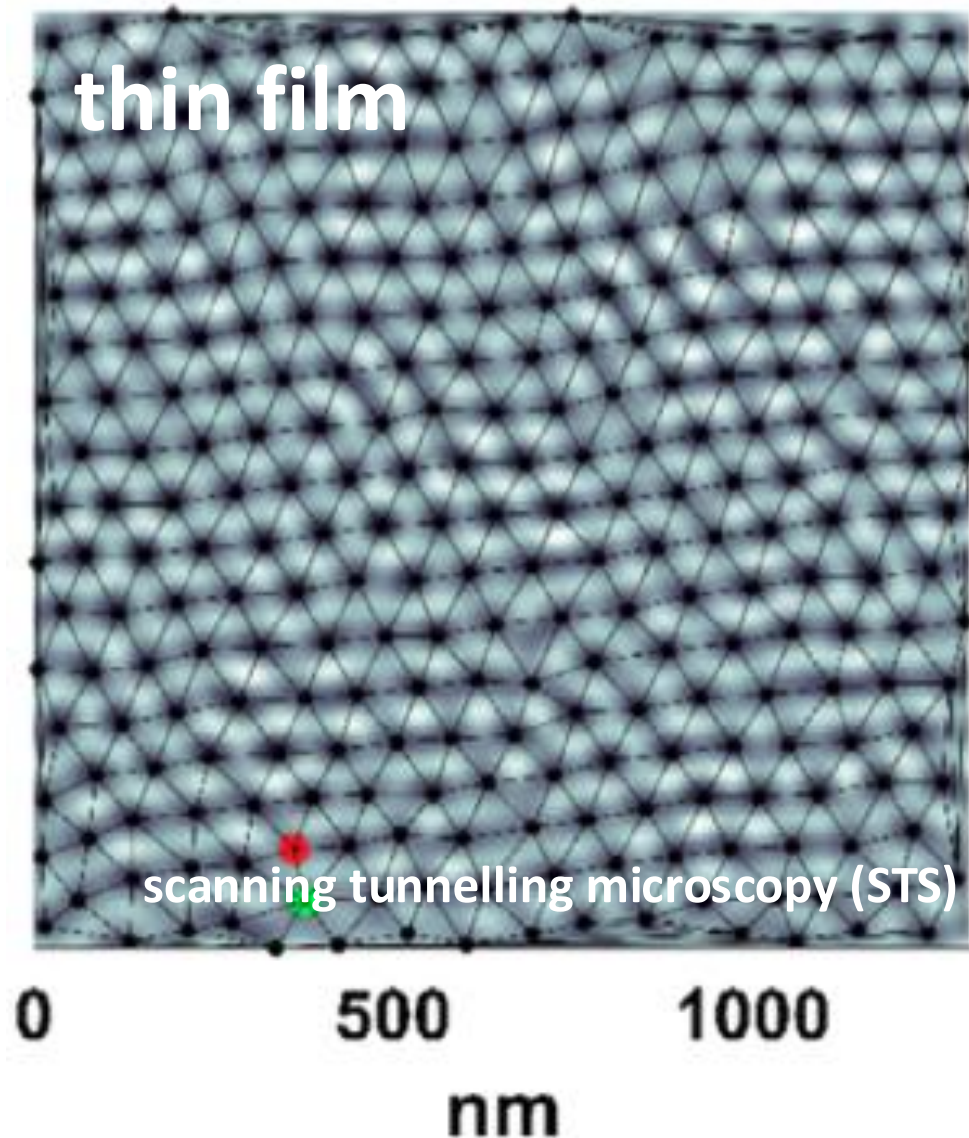
TRIAD “THEORY, SIMULATIONS ED EXPERIMENTS”



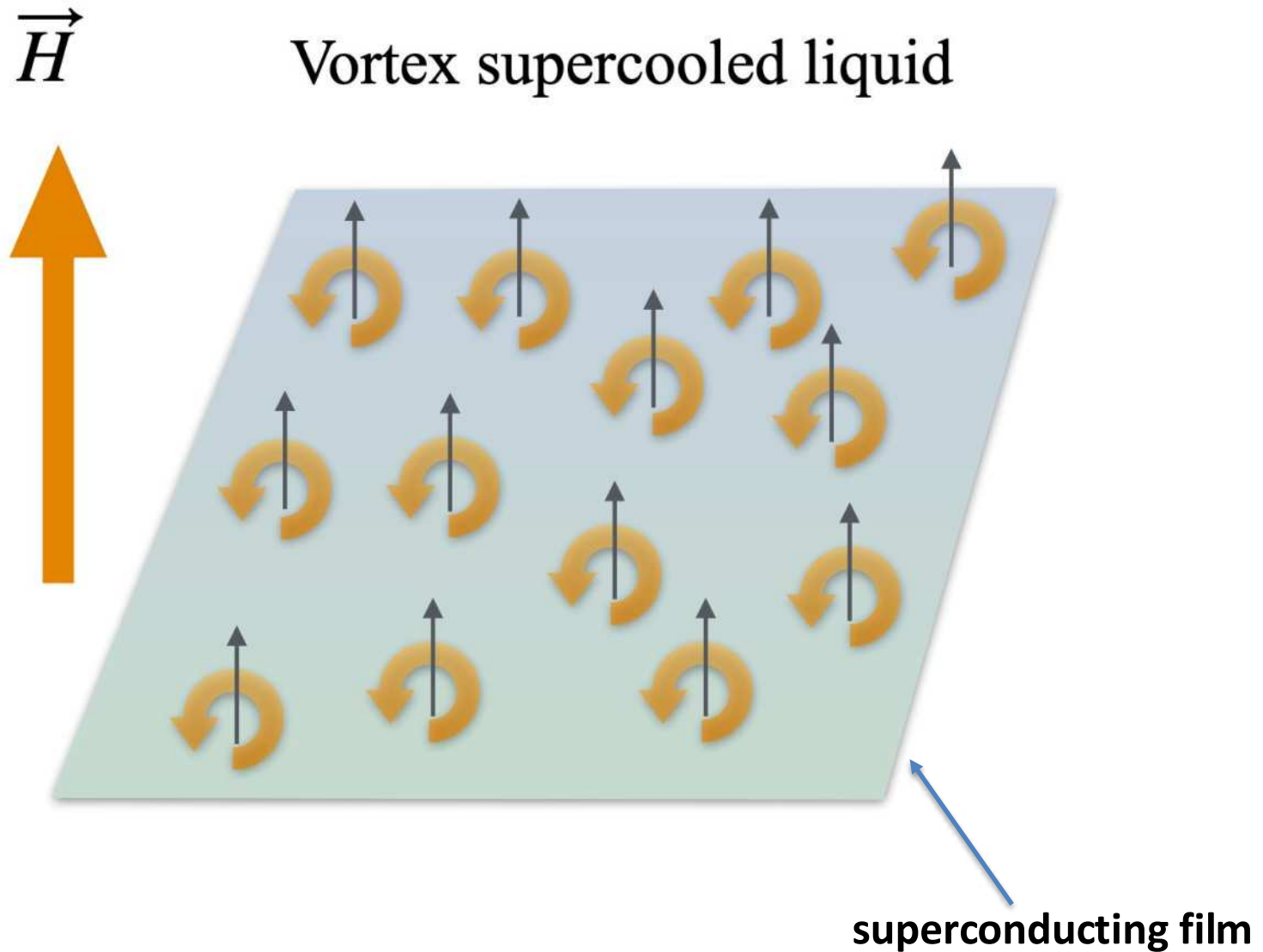
OUTLINE

- BRIEF HISTORY OF COMPUTERS
- SIMULATION OF COMPLEX SYSTEMS
- MODELS

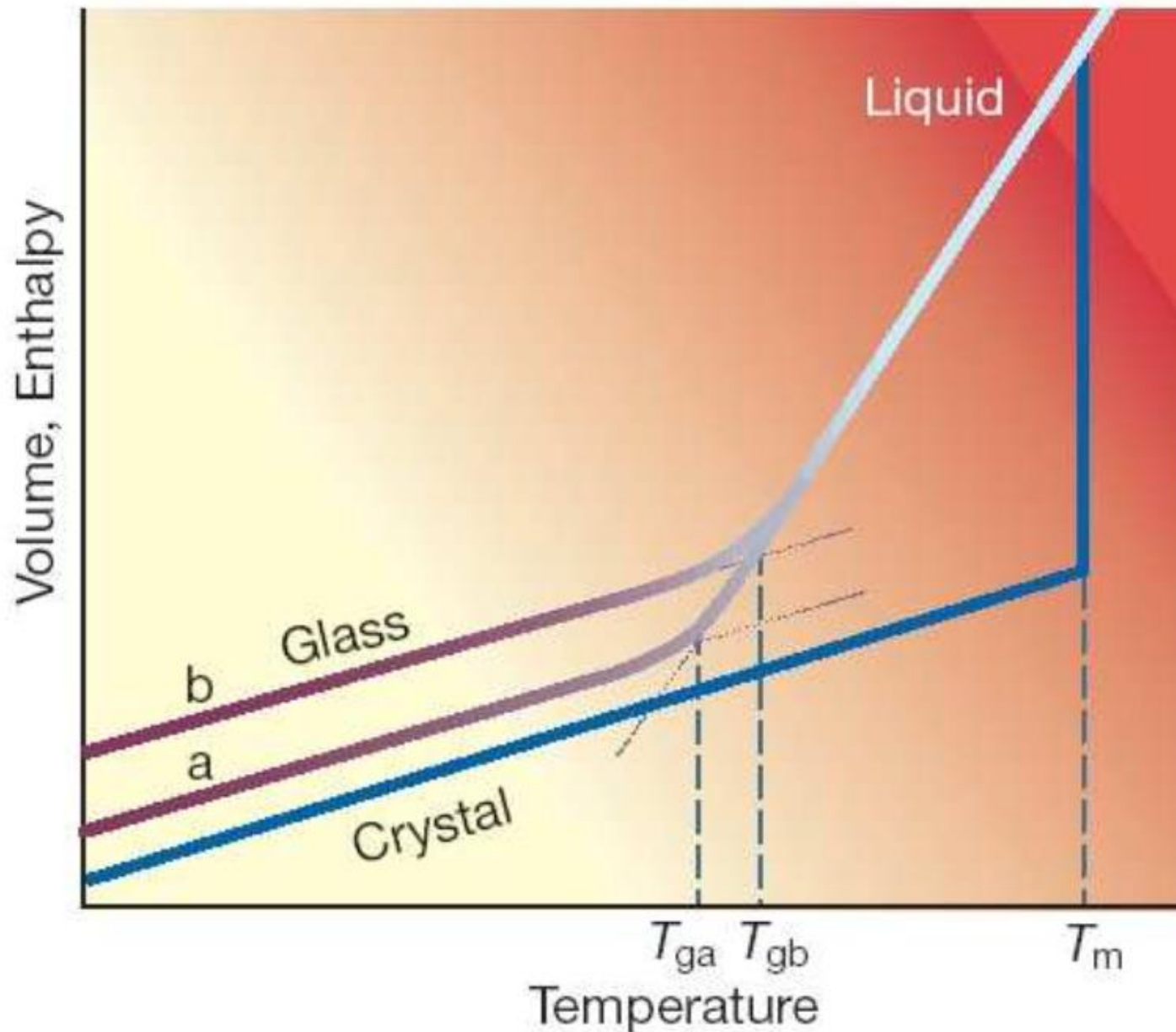
SUPERCONDUCTING FILMS



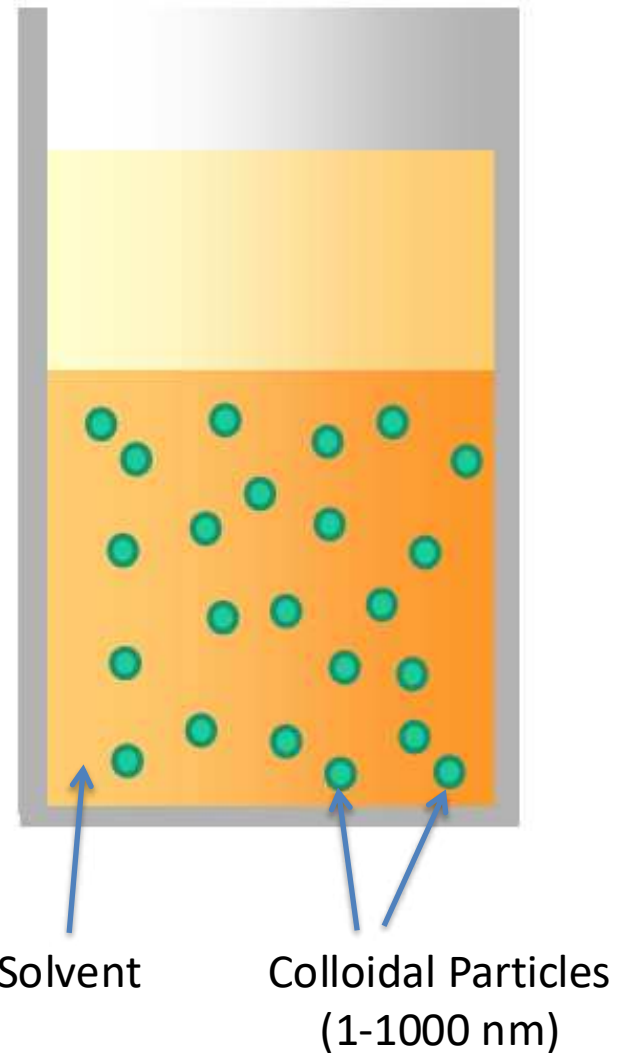
ABRIKOSOV VORTICES



SUPERCOOLED LIQUIDS AND GLASSES

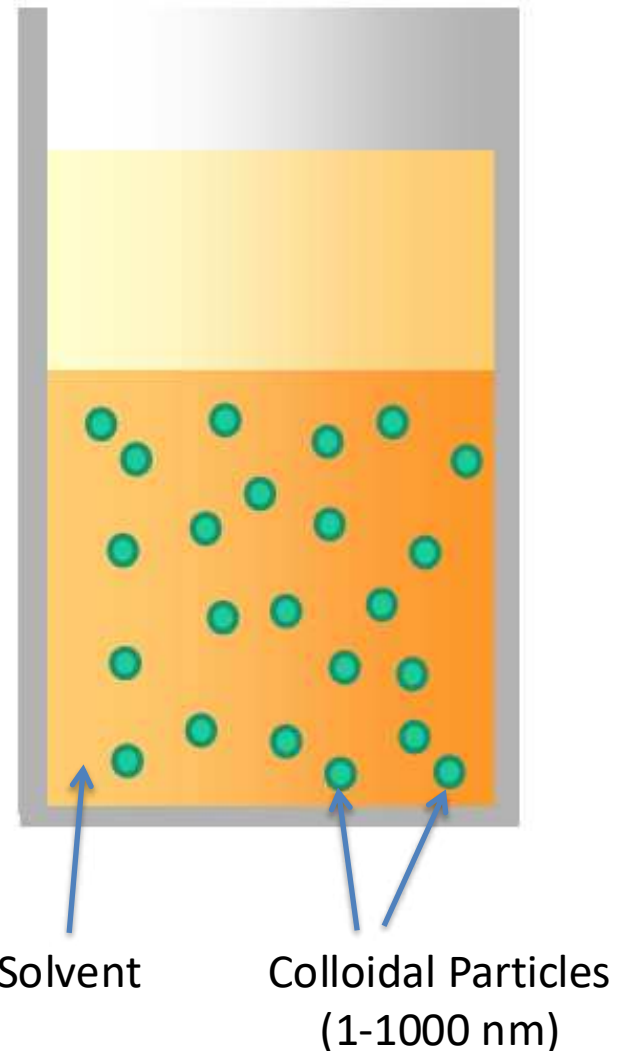


COLLOIDAL SYSTEMS



The word colloid comes from *kolla* and *eidos*
Kolla means glue and *eidos* means *type of*
Colloid means “type of glue” or “sticky stuff”

COLLOIDAL SYSTEMS

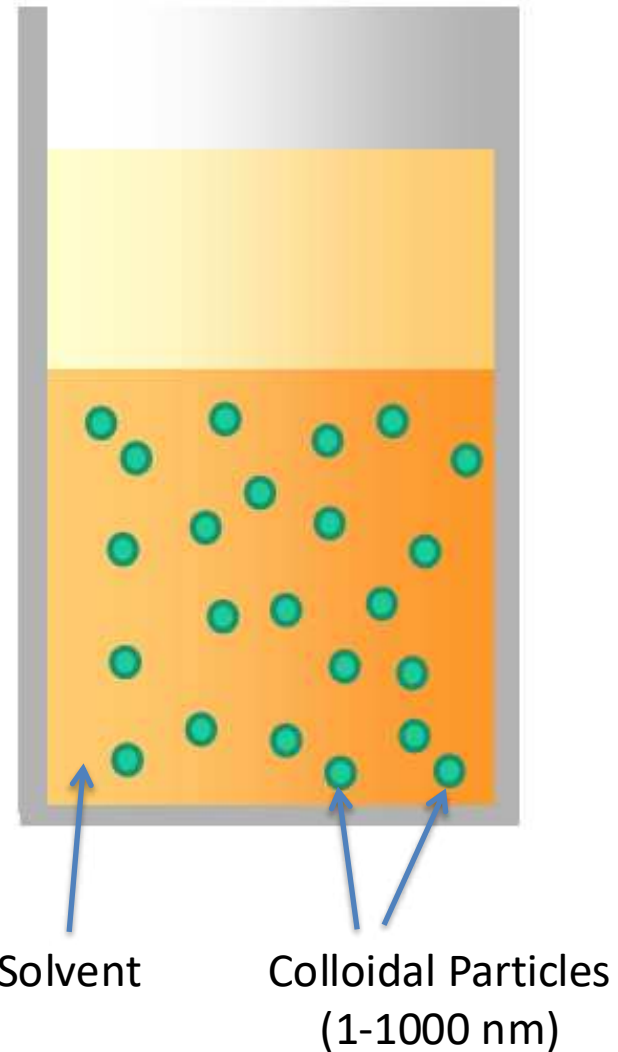


Solvent

Colloidal Particles
(1-1000 nm)

Examples of colloids: milk, fire-extinguisher foam, insulating foam, mayonnaise, blood, ink, cheeses, yoghurt, paints, ice-cream, suspensions of DNA duplexes and self-organized micelles

COLLOIDAL SYSTEMS



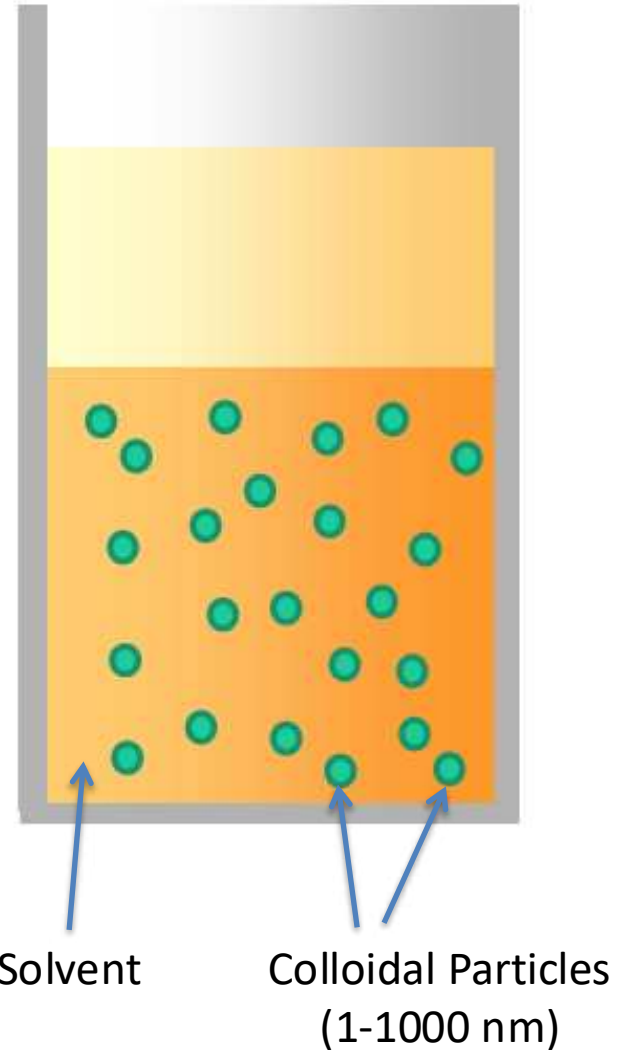
$$\Lambda = \sqrt{\frac{2\pi\hbar^2}{mk_B T}}$$

$$m \approx 10000 Da$$

$$T = 300 K$$

$$\Lambda \approx 0.4 pm^{34}$$

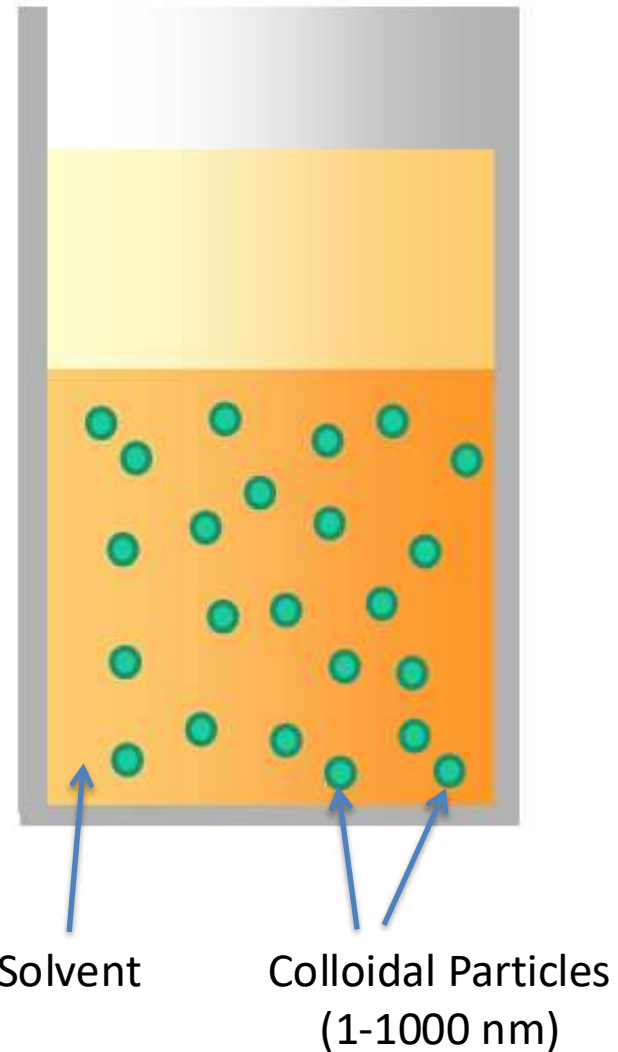
COLLOIDAL SYSTEMS



$$\Lambda < \frac{\{\text{colloid size}\}}{1000}$$

$$\Lambda \ll 1 - 1000 \text{ nm}$$

COLLOIDAL SYSTEMS



$$\Lambda \ll 1 - 1000 \text{ nm} \rightarrow$$

classical systems
not quantum-mechanical

A CHOICE, NOT A LIMITATION

classical simulations of mesoscopic length scales

Thermal de Broglie wavelength

$$\Lambda = \frac{h}{\sqrt{2\pi m k_B T}} \quad \text{compare with the interparticle distance } d$$

particle	Λ	regime
electron (300 K)	43 Å	quantum
H in water (300 K)	1.0 Å	≈ O–H bond: nuclear quantum effects
liquid He (4 K)	4.4 Å	quantum
liquid Ne (25 K)	0.78 Å	borderline
liquid Ar (85 K)	0.30 Å	classical ($\sigma = 3.4$ Å)
lysozyme (300 K)	0.008 Å	classical
colloid, $R = 100$ nm	1.4 fm	classical ($d \sim \mu\text{m}$)

$\Lambda \gtrsim d$ (light atoms, low T): path-integral MD/MC, affordable today with ML potentials (e.g. DeePMD) — still a classical simulation, of ring polymers!

Length scales and methods

10^{-10} m electrons, chemical bonds
quantum: DFT, ab initio MD



ML potentials: DFT accuracy at force-field cost

10^{-9} m atoms, small molecules
classical all-atom force fields



ML-CG: force matching of the PMF

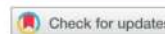
10^{-8} – 10^{-6} m colloids, biomacromolecules, liquid crystals, soft matter
coarse-grained models (also ML-CG) — **THIS COURSE**

$> 10^{-5}$ m fluids and solids as continua
hydrodynamics, elasticity

Classical is not an approximation here: it is the right physics at these scales.

Quantum mechanics enters through the interactions: ML potentials carry electronic (and, with path integrals, nuclear) quantum effects up to classical, mesoscopic simulations.

MOLECULAR PHYSICS e2364834
<https://doi.org/10.1080/00268976.2024.2364834>

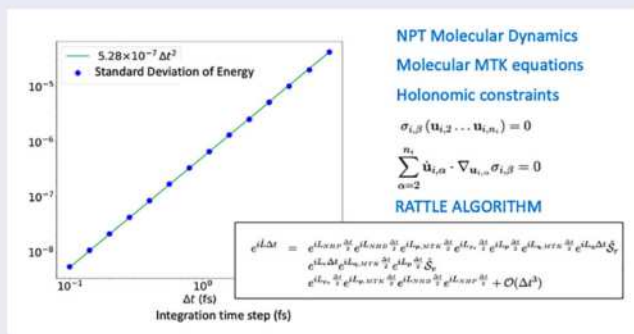
C. De Michele^a and G. Ciccotti^{b*}

^aDepartment of Physics, "Sapienza" Università di Roma, Rome, Italy; ^bIstituto per le Applicazioni del Calcolo CNR, Rome, Italy

For generating an isobaric–isothermal ensemble in molecular dynamics simulations of atomic systems a correct ensemble distribution can be obtained by the approach of Martyna, Tobias and Klein [J. Chem. Phys. **101**, 4177–4189 (1994)]. The constituting equations of the latter approach have been also generalised to molecular systems [M. E. Tuckerman, *Statistical Mechanics: Theory and Molecular Simulation*, Oxford Graduate Texts (2010)] using the molecular virial instead of the atomic one. An isothermal–isobaric method for systems with holonomic molecular constraints has been also introduced in the past [G. Kalibaeva, M. Ferrario and G. Ciccotti, Mol. Phys. **101** 765–778 (2003)] but the factorisation was not worked out completely, resulting in a non-reversible first-order algorithm. Here, we propose the correct factorisation and implement a reversible integrator based on Martyna, Tobias and Klein equations for a system of molecules subjected to holonomic constraints. We add to the algorithm the Suzuki–Yoshida treatment that improves the energy conservation or permits to go to larger time steps. Finally, we test our algorithm by applying it to the dynamics of a *Orthoterphenyl* model.

Received 28 November 2023
Accepted 31 May 2024

Molecular dynamics;
reversible integrator;
measure-preserving
algorithm;
isothermal–isobaric
ensemble; holonomic
constraints

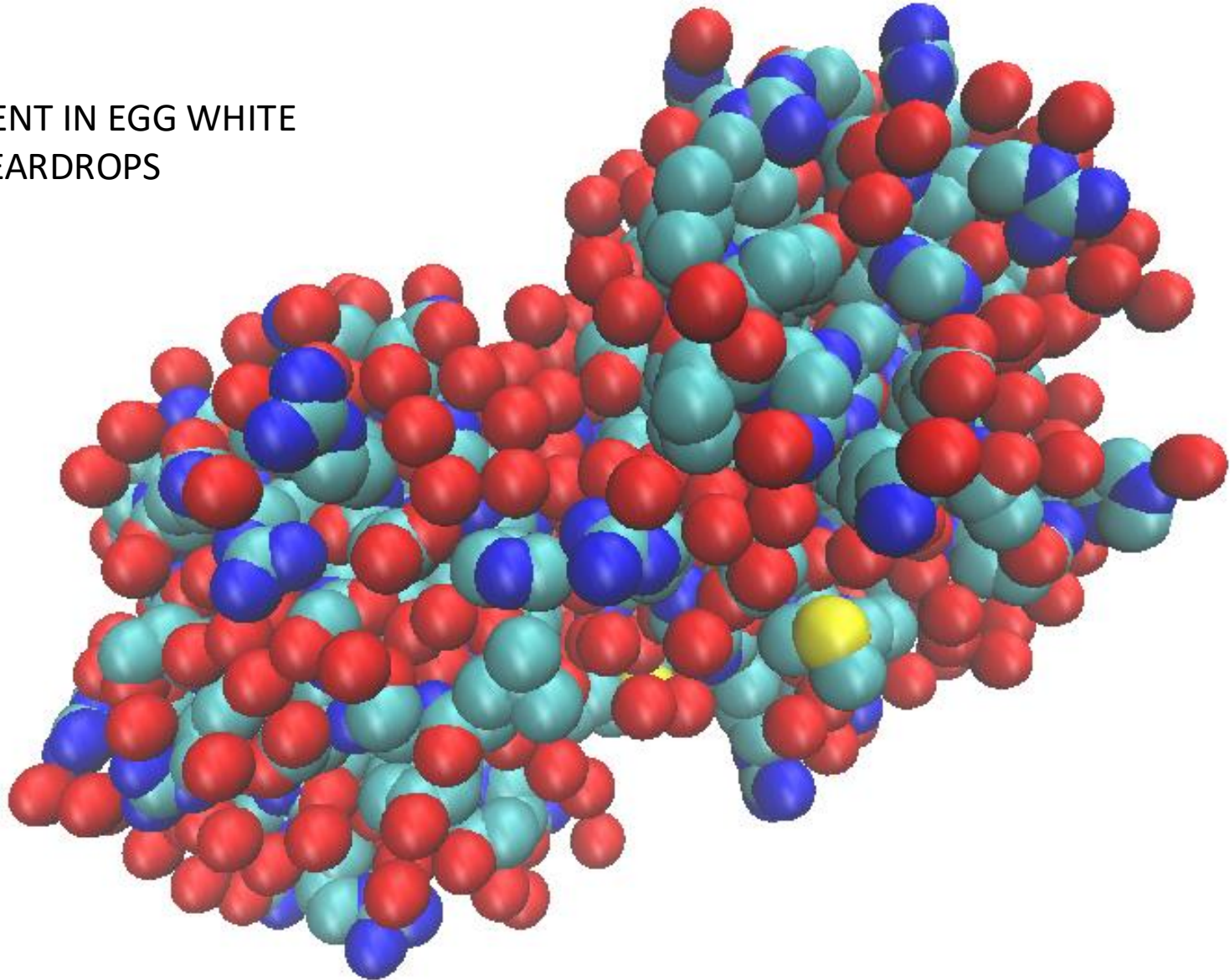


OUTLINE

- BRIEF HISTORY OF COMPUTERS
- SIMULATION OF COMPLEX SYSTEMS
- **MODELS**

PROTEIN (LYSOZYME)

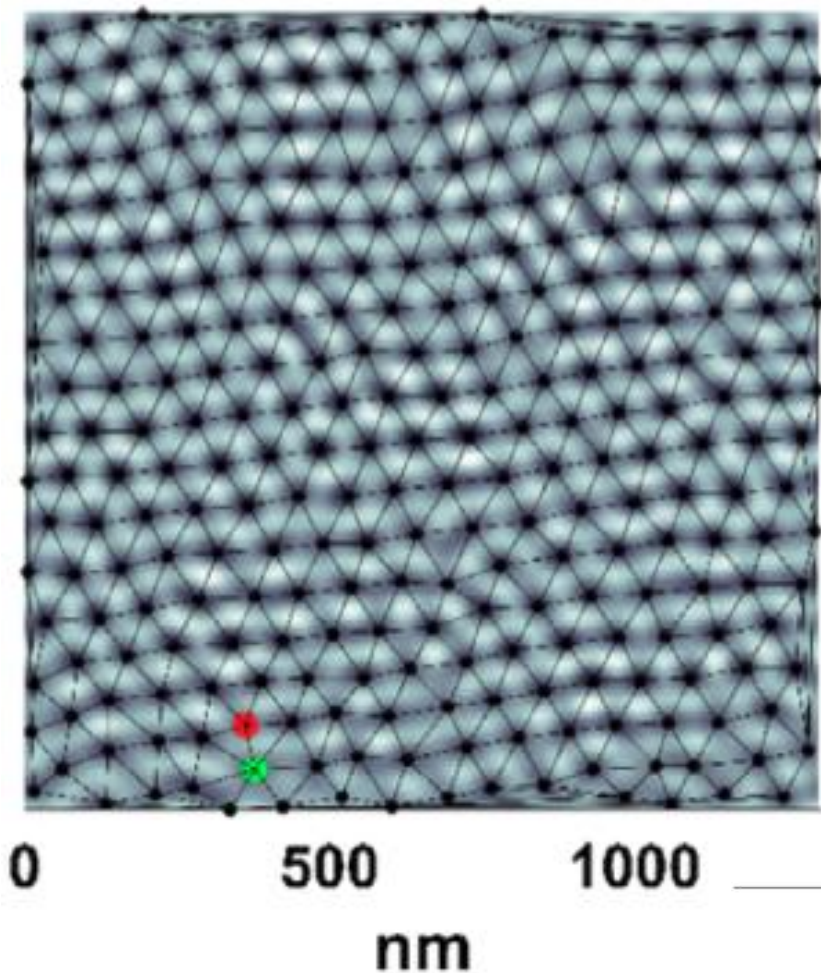
PRESENT IN EGG WHITE
OR TEARDROPS



NOBEL PRIZE FOR CHEMISTRY 2013: LEVITT, KARPLUS, WARSHEL

MODELLING THIN SC FILMS

EXPERIMENT

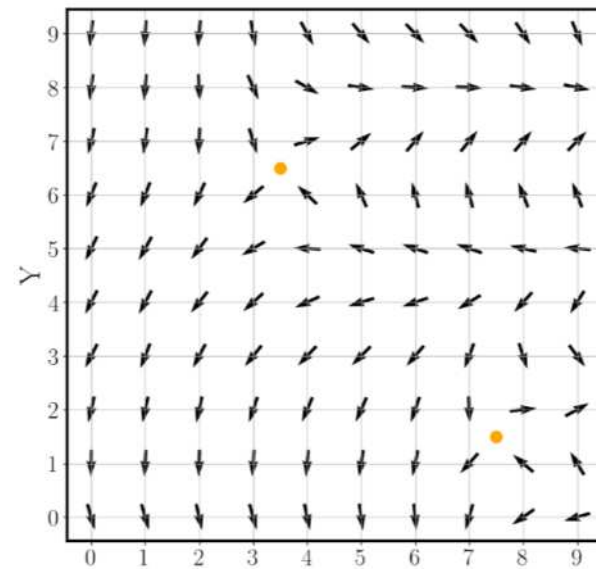


XY-MODEL

$$H_{XY} = - \sum_{i,\mu=\hat{x},\hat{y}} J_i^\mu \cos(\theta_i - \theta_{i+\mu} + F_i^\mu).$$

Effective spin model

coupling to
external field



Theory

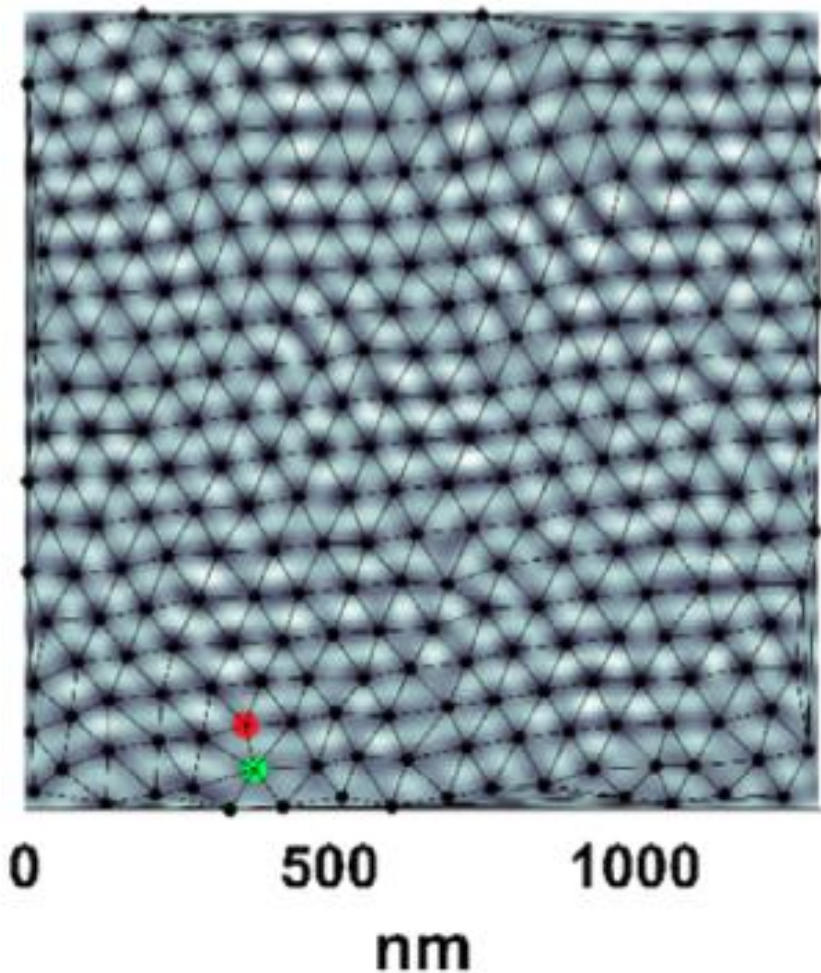
PHYSICAL REVIEW B **107**, 014509 (2023)

Transport signatures of fragile glass dynamics in the melting of the two-dimensional vortex lattice

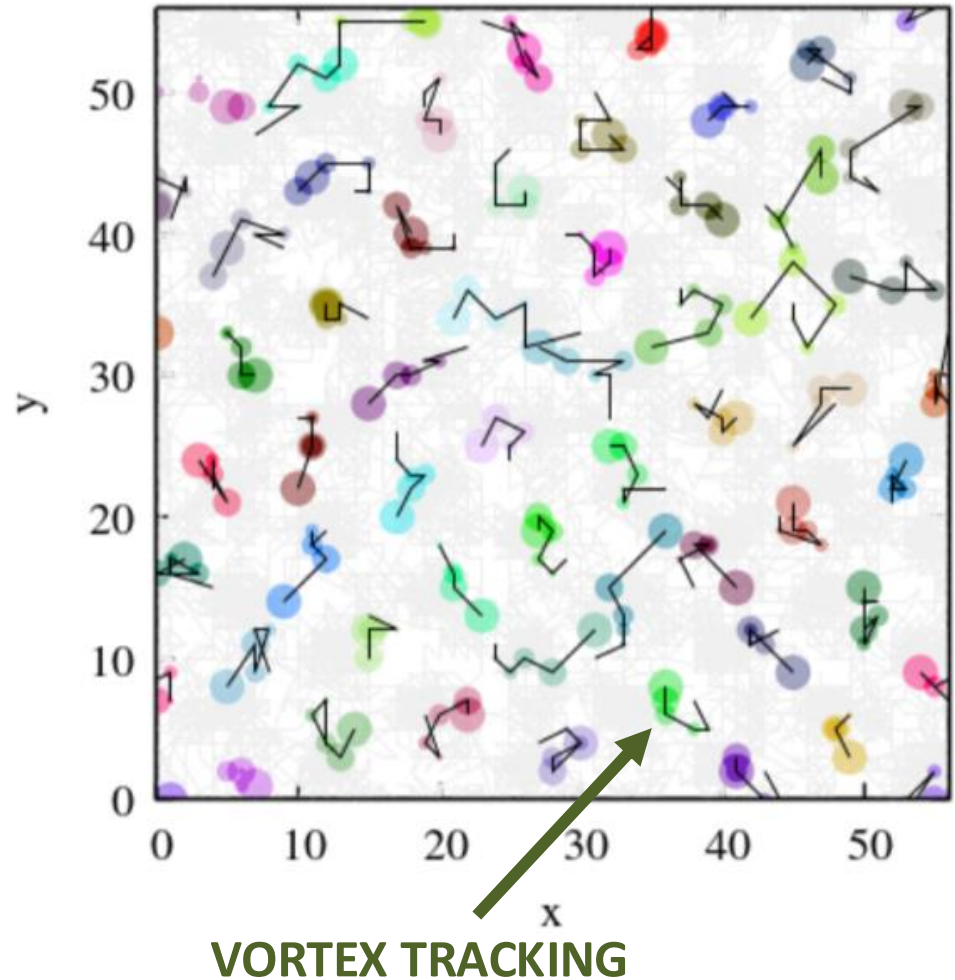
I. Maccari^{1,*}, Bal K. Pokharel^{2,3,*}, J. Terzic^{2,4}, Surajit Dutta⁵, J. Jesudasan⁵, Pratap Raychaudhuri⁵, J. Lorenzana⁶,
C. De Michele⁷, C. Castellani^{7,6}, L. Benfatto^{7,6,8} and Dragana Popović^{2,3,9}

DIFFUSING VORTICES

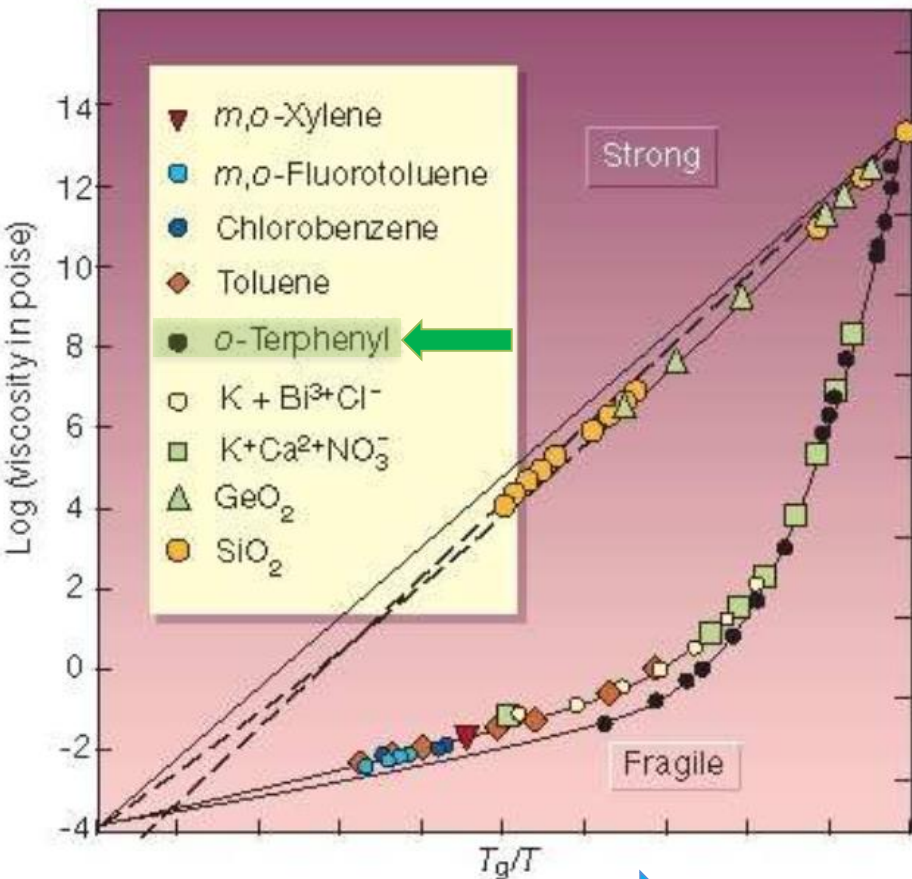
EXPERIMENT



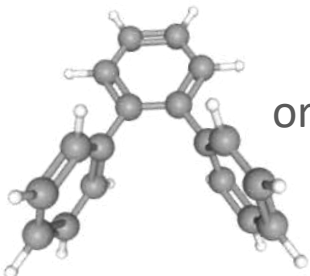
XY-MODEL



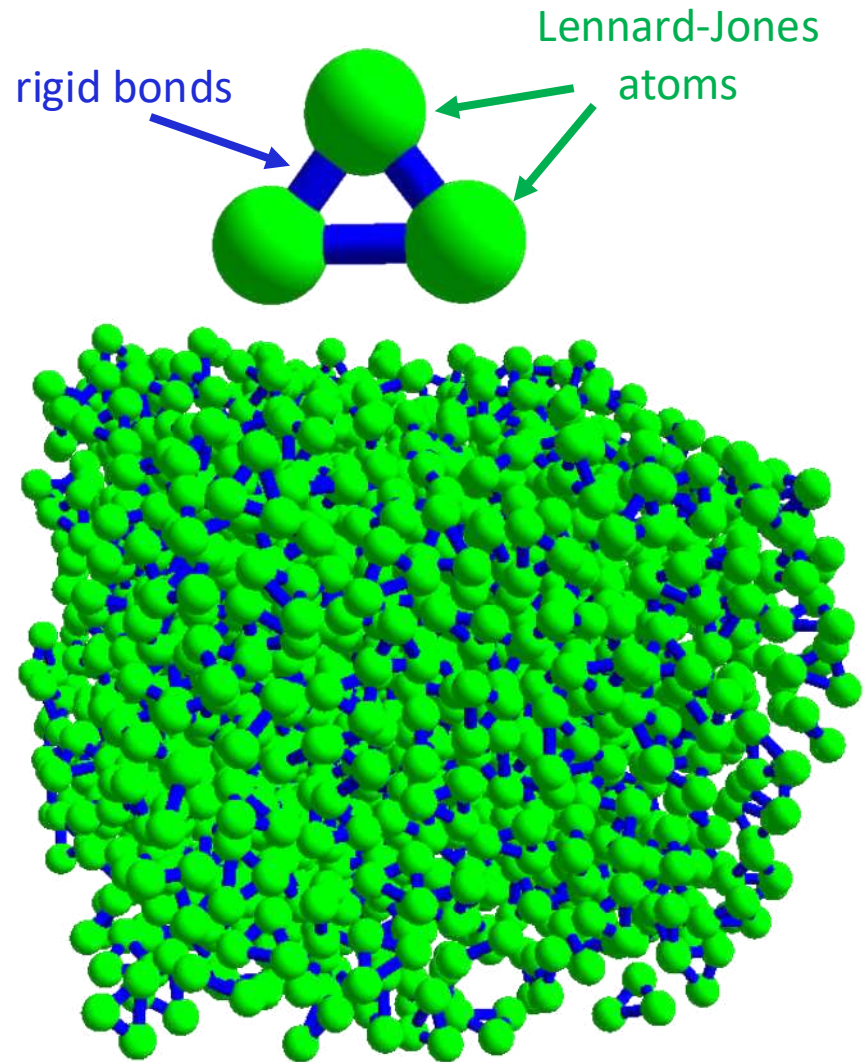
A MOLECULAR GLASS: ORTHOTERPHENYL



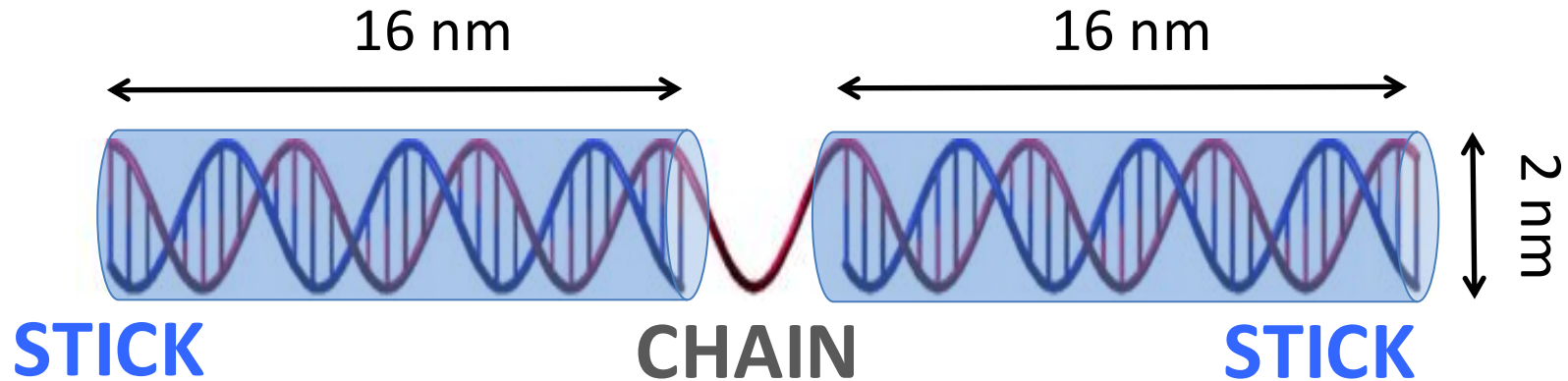
decreasing temperature



orthoterphenyl



DNA CHAIN-STICK MODEL

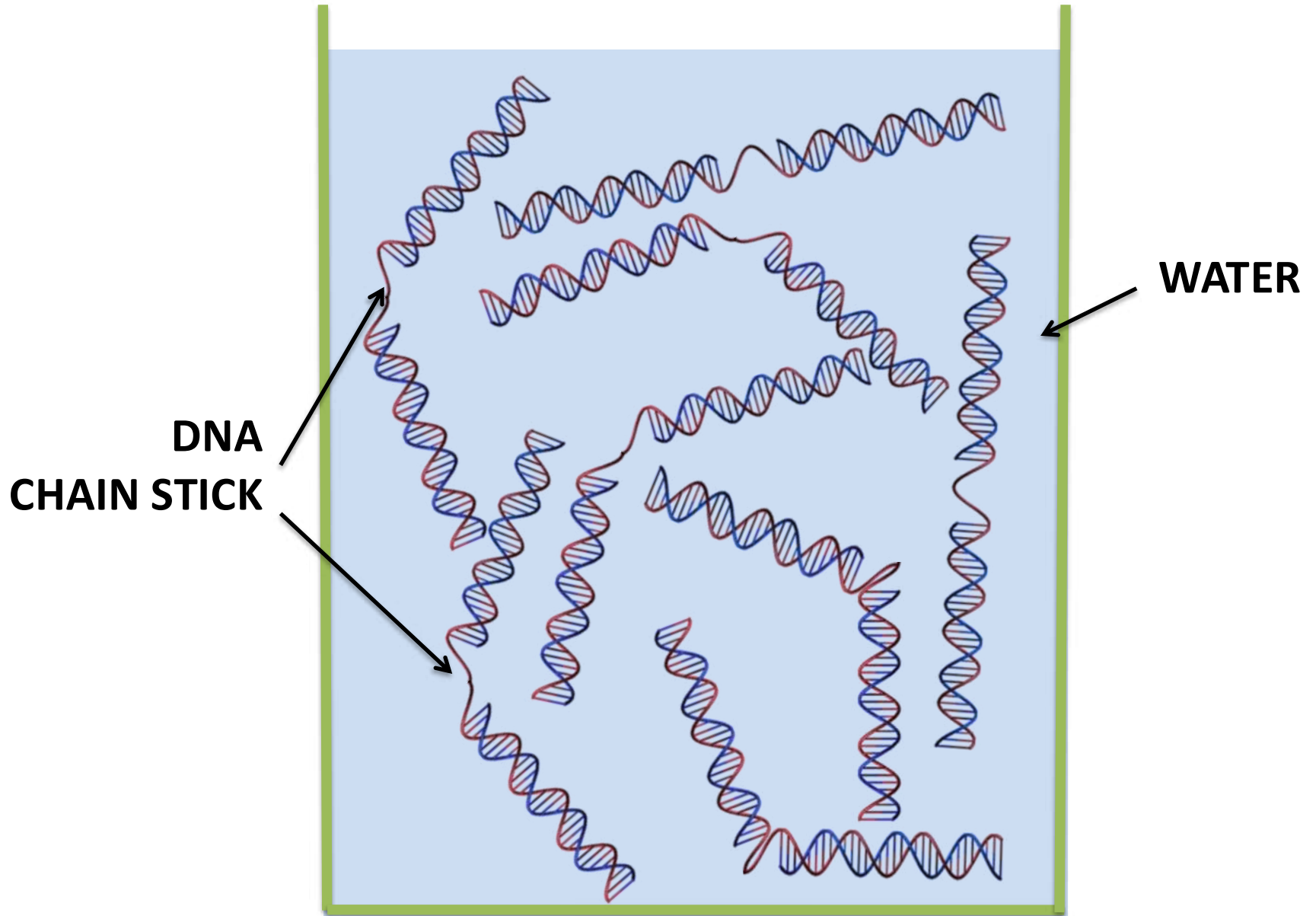


Smectic phase in suspensions of gapped DNA duplexes

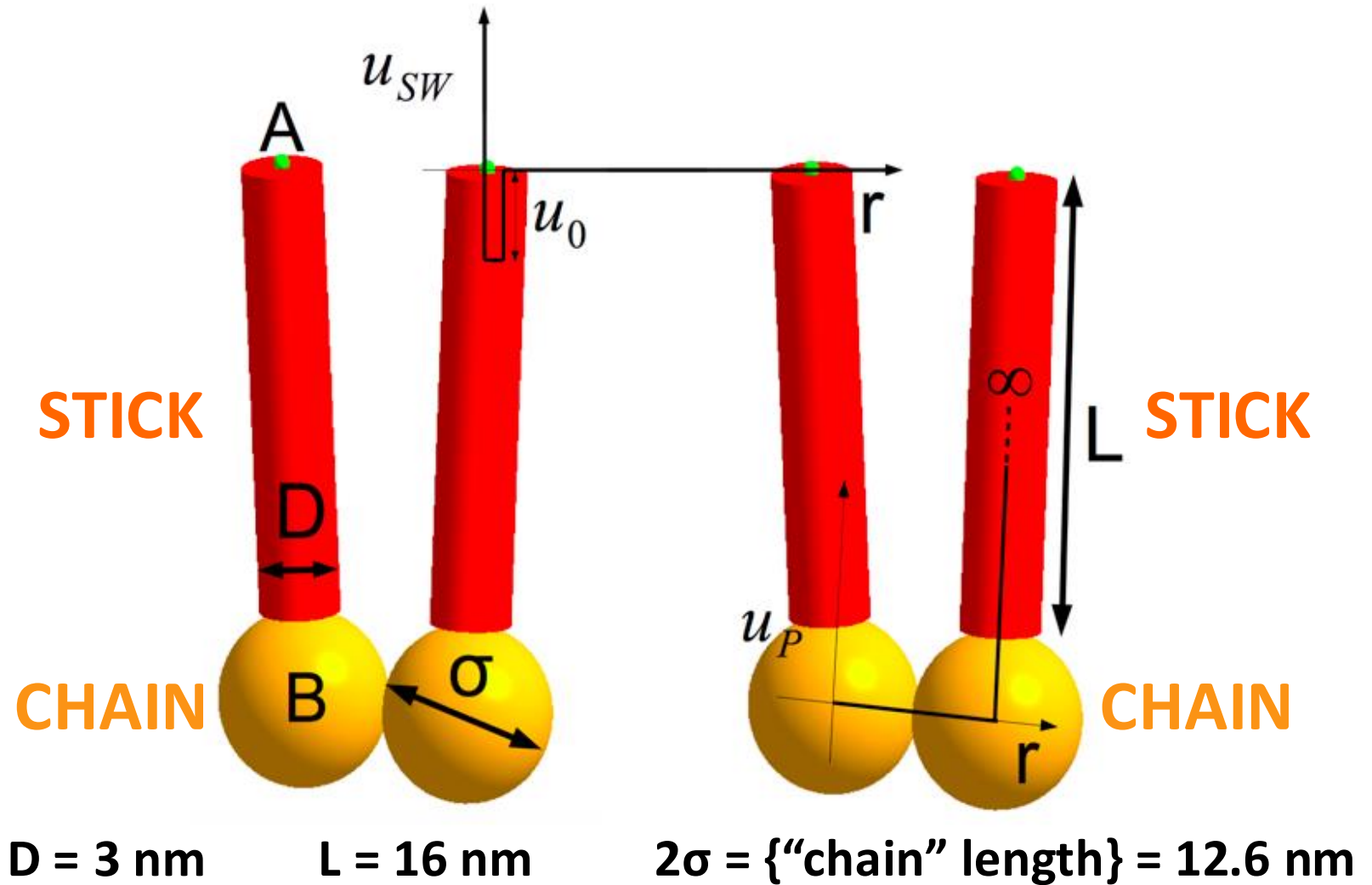
Mirosław Salamonczyk, Jing Zhang, Giuseppe Portale, Chenhui Zhu, Emmanuel Kentzinger, James T. Gleeson, Antal Jakli, **Cristiano De Michele**✉, Jan K. G. Dhont, Samuel Sprunt & **Emmanuel Stiakakis**✉



EXPERIMENTAL SYSTEM



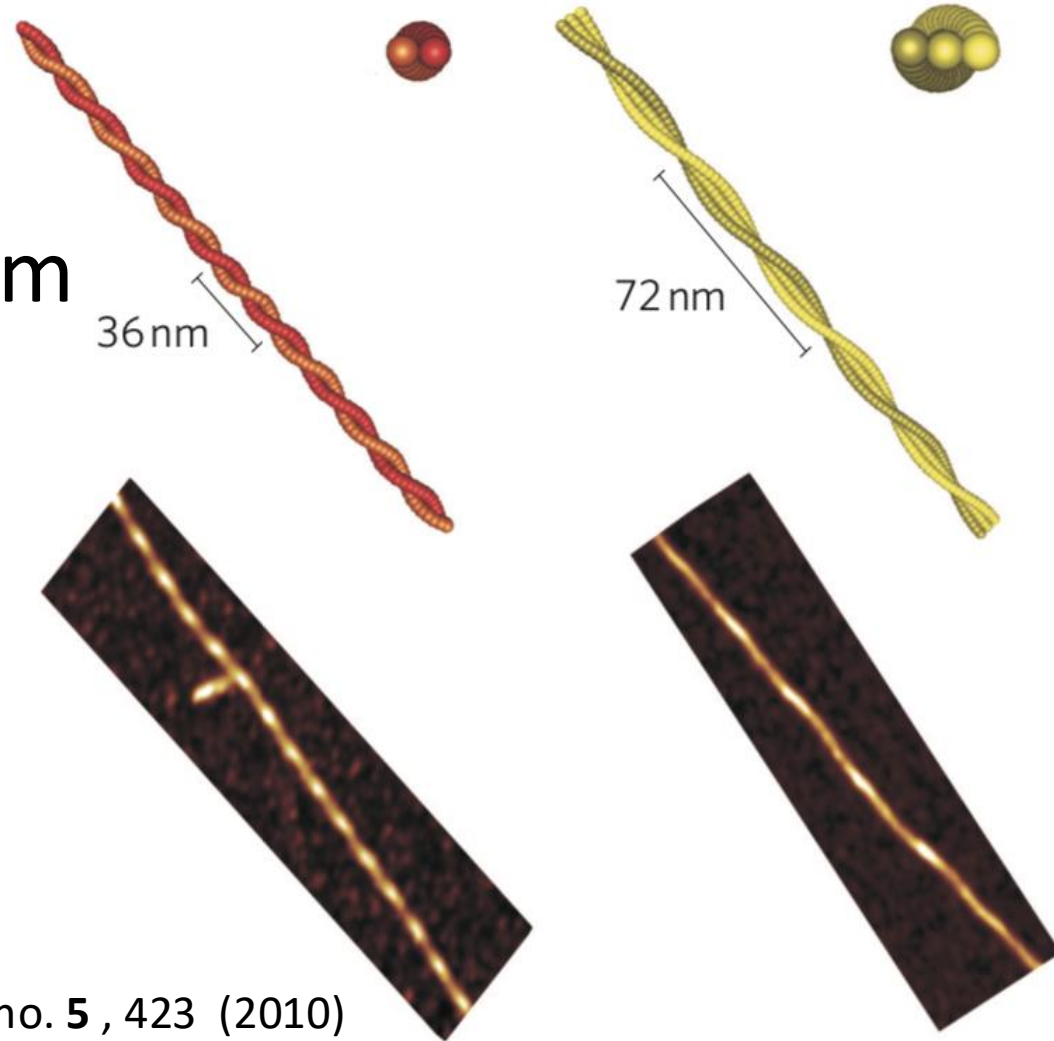
COARSE-GRAINED MODEL



AMYLOID FIBRILS

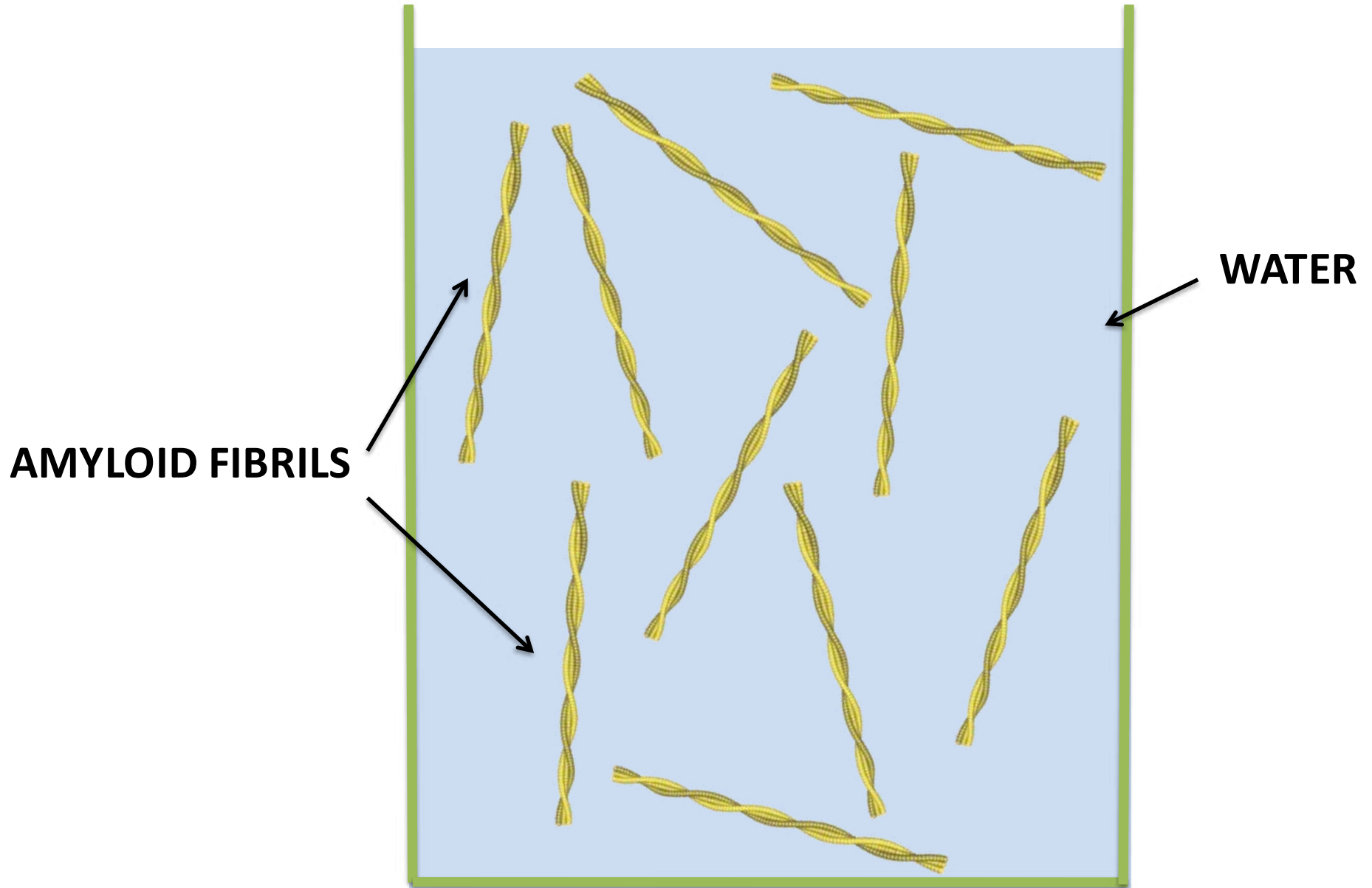
B-lactoglobulin

$\langle L \rangle \approx 1\text{-}20\ \mu\text{m}$

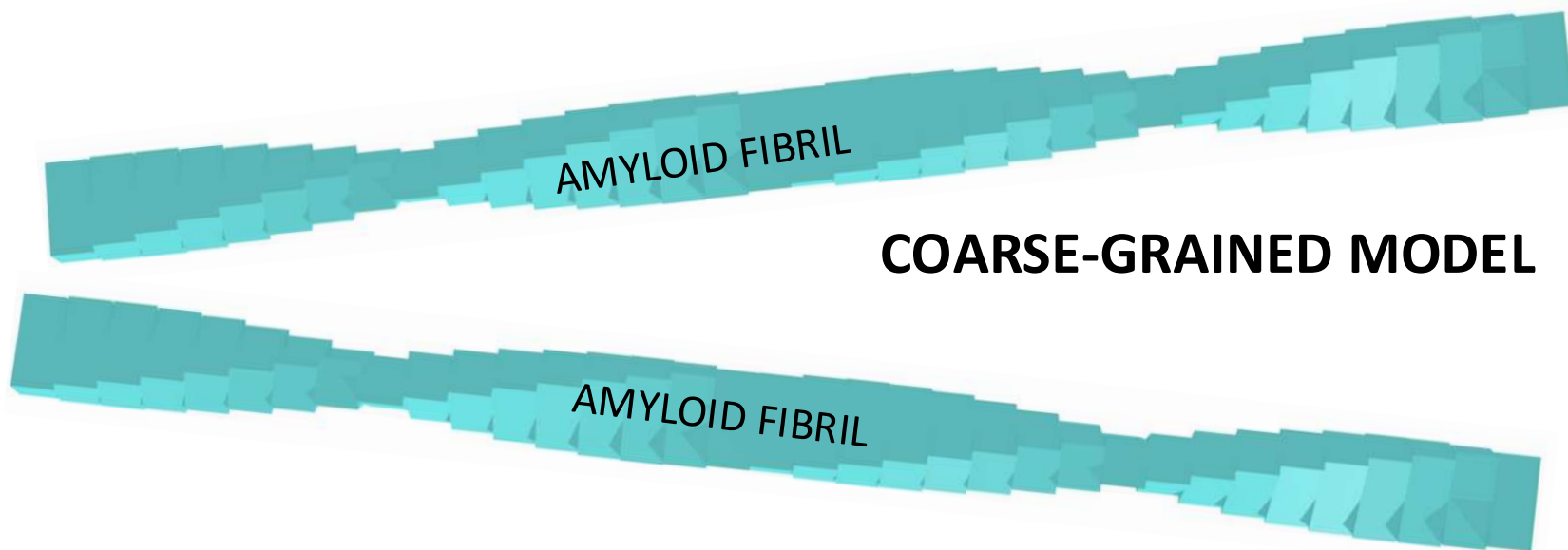


J. Adamick *et al.* Nat. Nano. **5**, 423 (2010)

AMYLOID FIBRILS: EXPERIMENTAL SYSTEM



AMYLOID FIBRILS: COARSE-GRAINED MODEL



ACS NANO

Cite This: ACS Nano XXXX, XXX, XXX–XXX

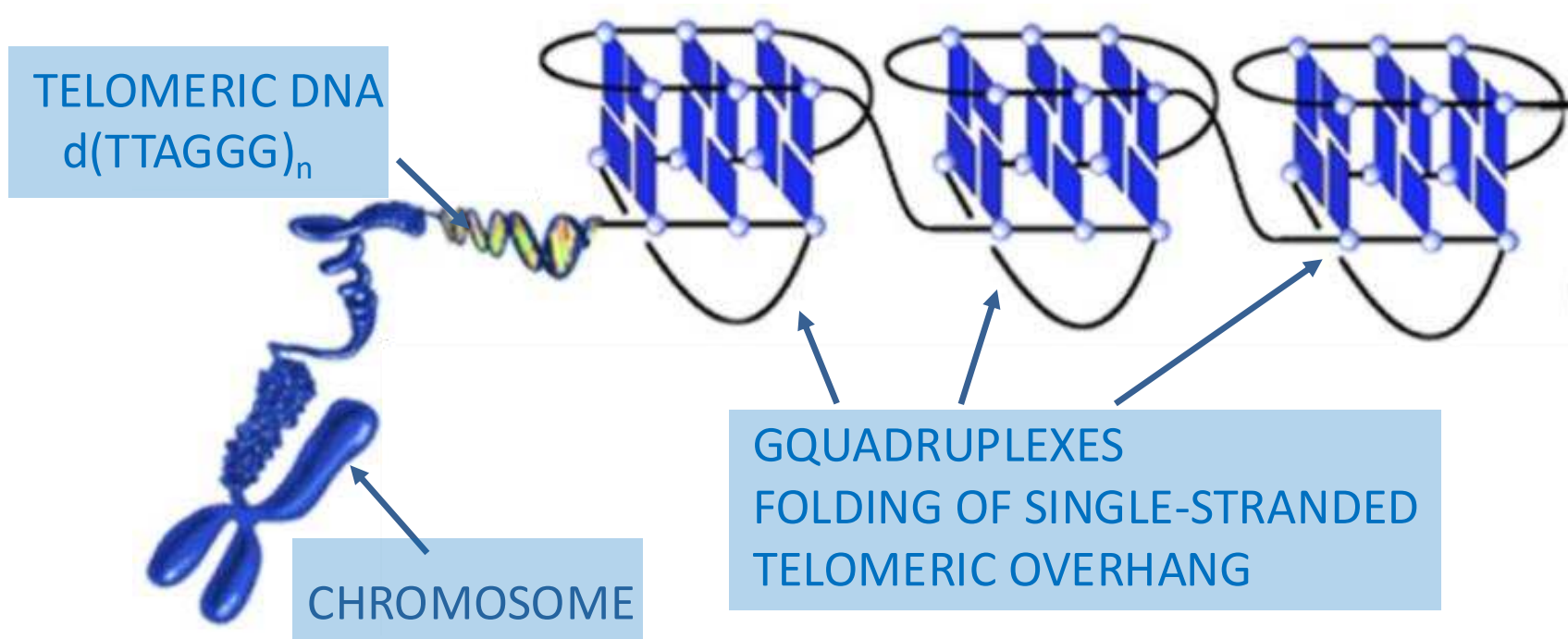
www.acsnano.org

Amyloid Fibrils Length Controls Shape and Structure of Nematic and Cholesteric Tactoids

Massimo Bagnani,[†] Gustav Nyström,[†] Cristiano De Michele,[‡] and Raffaele Mezzenga^{*,†,§}

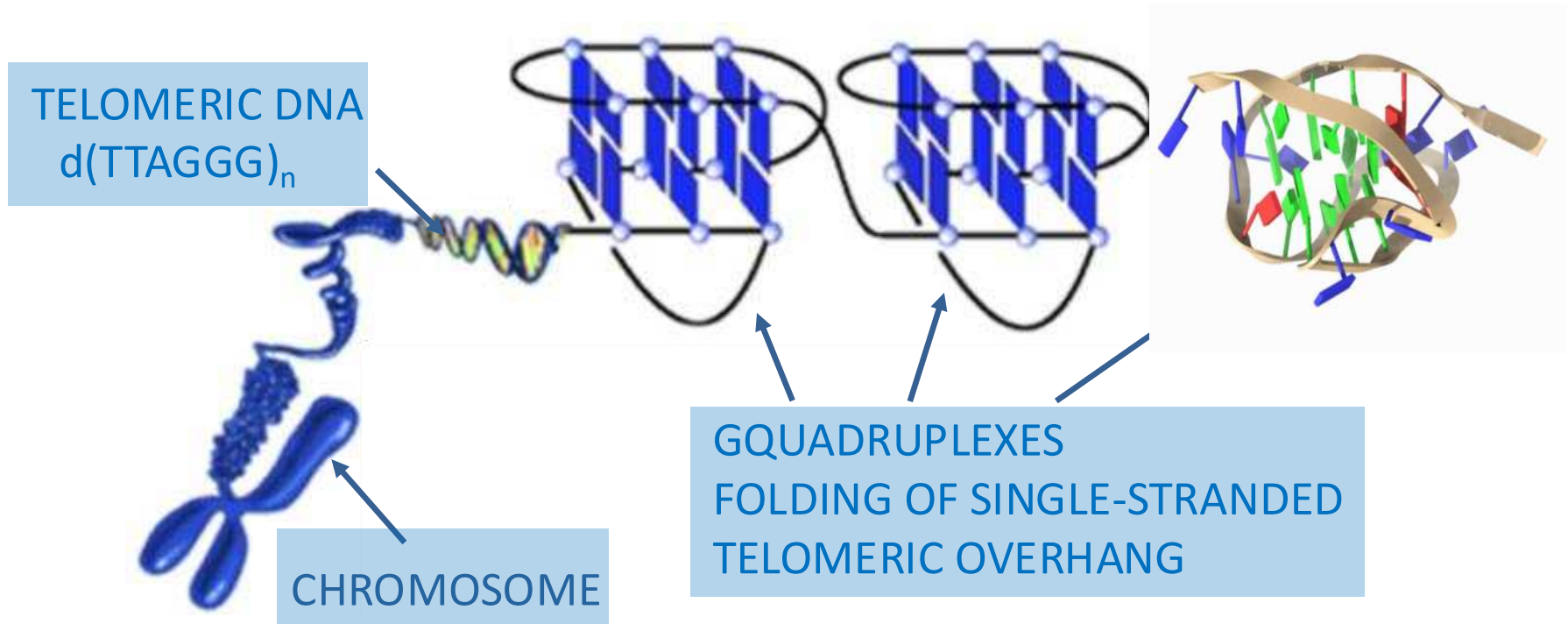
ARTICLE

TELOMERIC DNA: MULTIMERIC GQUADRUPLLEXES (G4s)



L. Petraccone, Top. Curr. Chem. **330**, 23 (2013)

TELOMERIC DNA: MULTIMERIC GQUADRUPLLEXES (G4s)



L. Petraccone, Top. Curr. Chem. **330**, 23 (2013)

TELOMERIC DNA: MULTIMERIC GQUADRUPLEXES (G4s)

TELOMERIC DNA
 $d(\text{TTAGGG})_n$

- In telomeres region up to 600 nucleotides
- Multimers with more than 10 G4s may form

Therapeutic Relevance

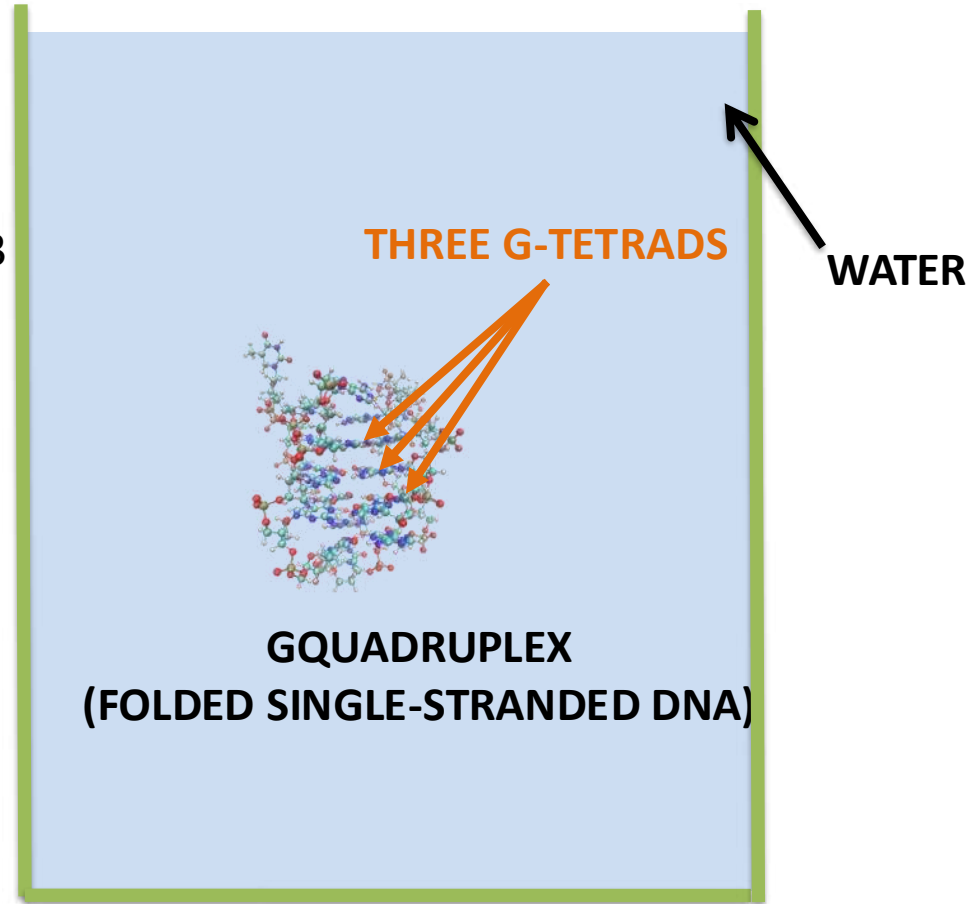
- In more than 85% of cancer cells telomeres do not shorten on replication
- Multimeric G4s in telomeres are targets for anticancer drugs

L. Petraccone, Top. Curr. Chem. **330**, 23 (2013)

FOLDED DNA FRAGMENTS (G-QUADRUPLEXES)

TEL22

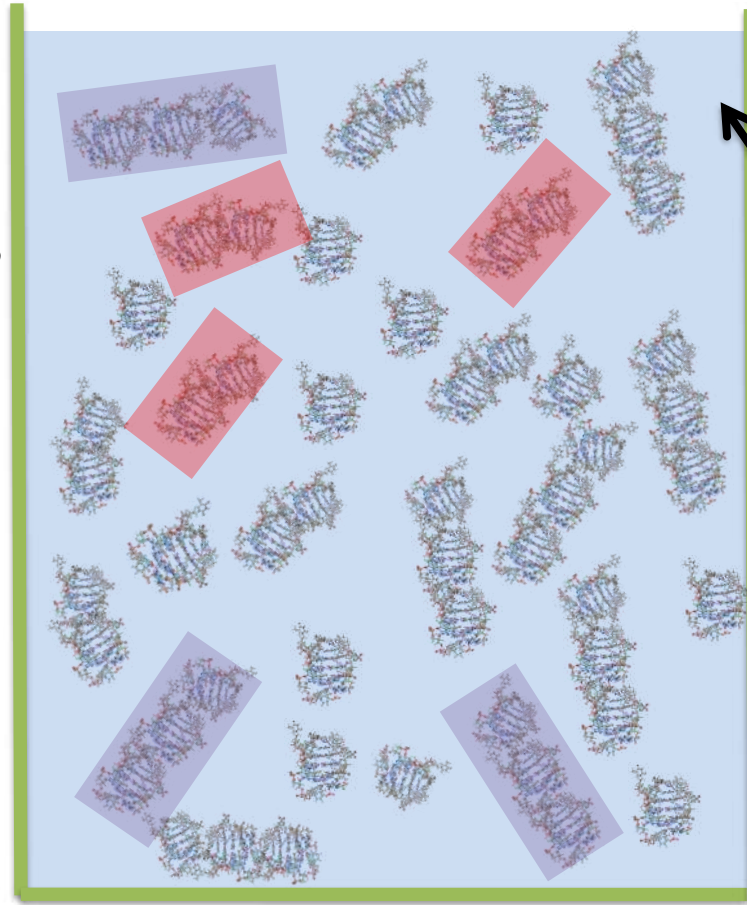
AGGG(TTAGGG)₃



REVERSIBLE AGGREGATION OF GQUADRUPLLEXES

TEL22

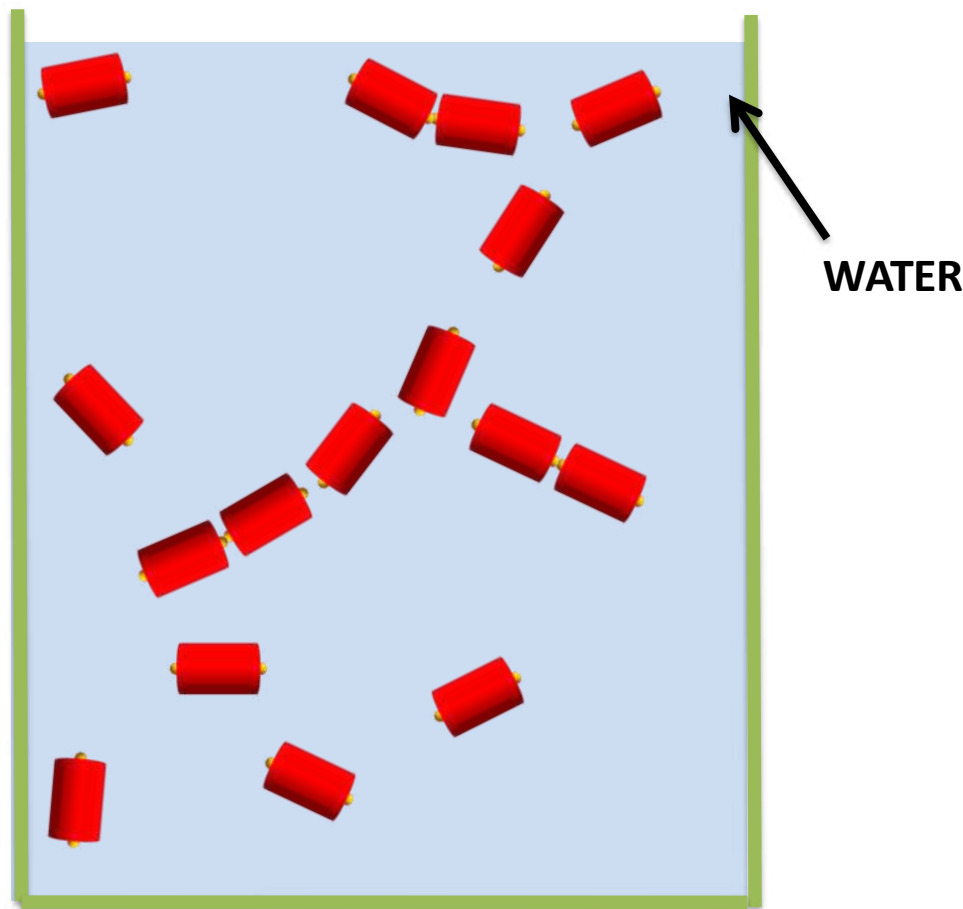
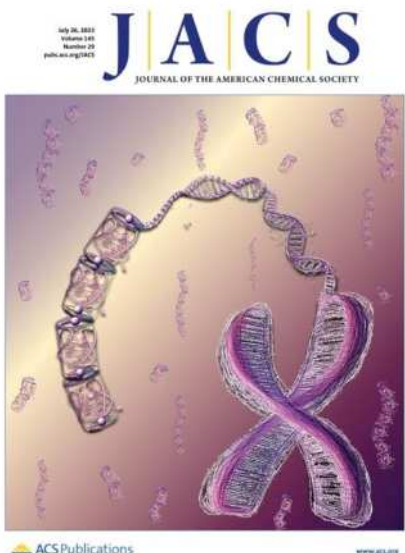
AGGG(TTAGGG)₃



WATER

STACKING OF GQUADRUPLLEXES
FORMING REVERSIBLE
DIMERS, **TRIMERS**, ...

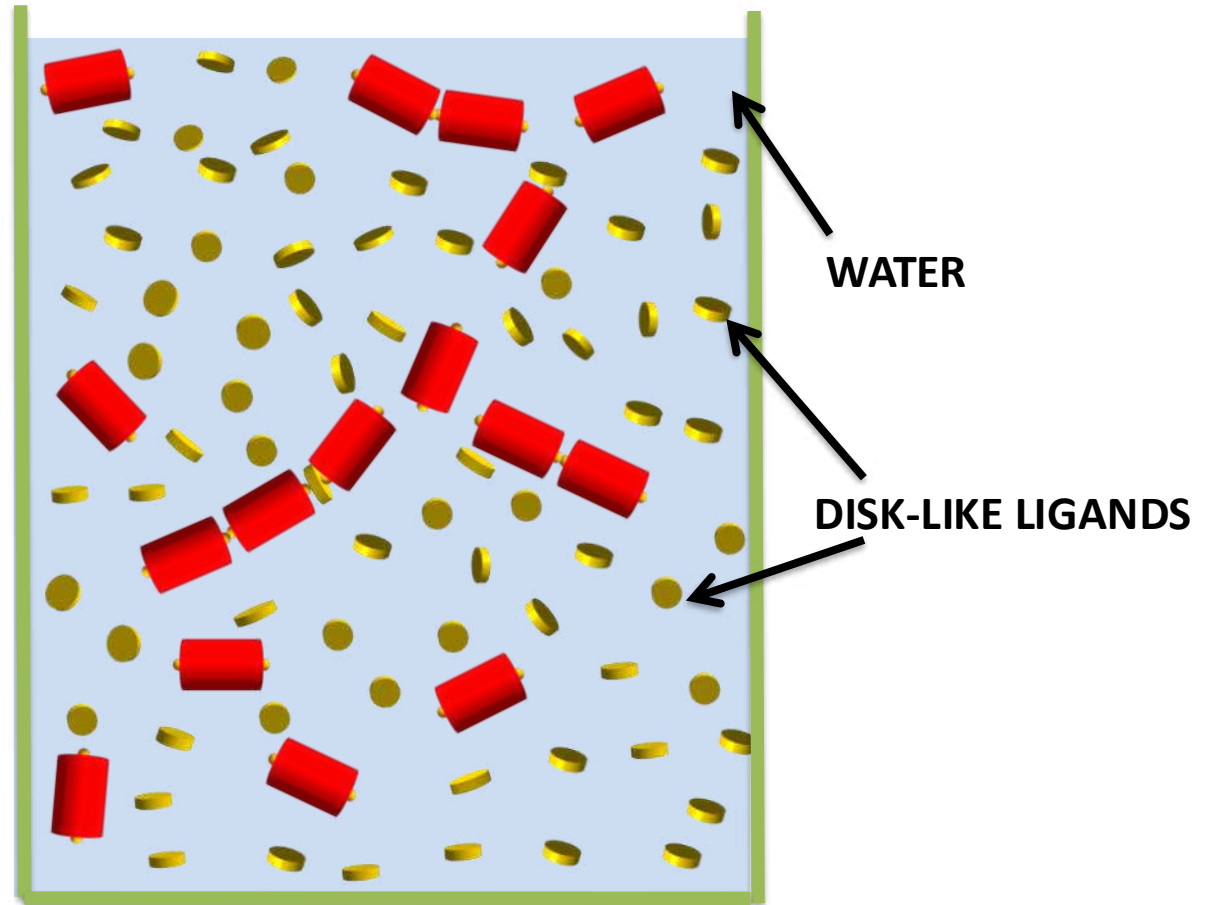
GQUADRUPLEX MODELING

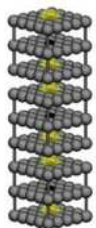


Stacking Interactions and Flexibility of Human Telomeric Multimers (2023)

Benedetta Petra Rosi, Valeria Libera, Luca Bertini, Andrea Orecchini, Silvia Corezzi, Giorgio Schirò, Petra Pernot, Ralf Biehl, Caterina Petrillo, Lucia Comez*, Cristiano De Michele*, and Alessandro Paciaroni*

SOLUTION OF MULTIMERS AND LIGANDS (ANTI-CANCER DRUGS)

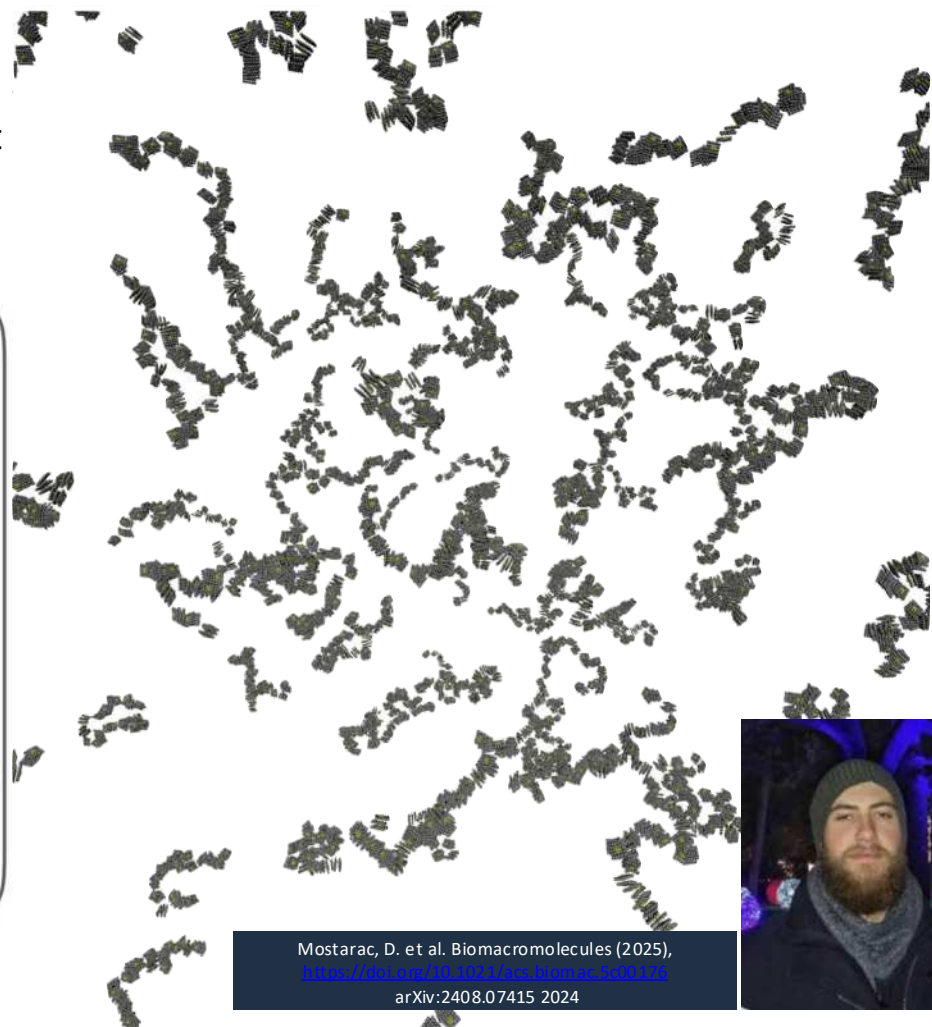
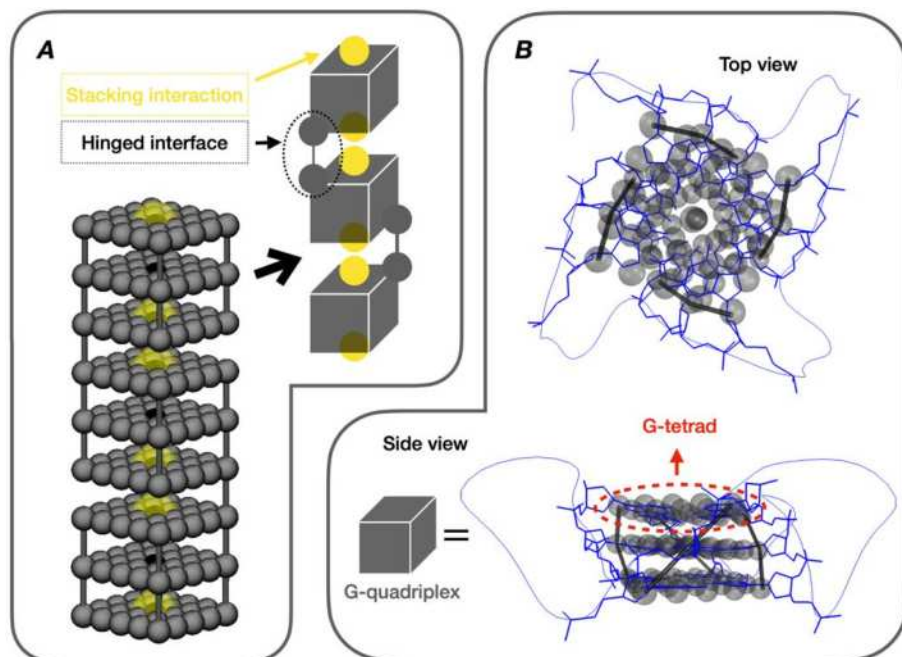




SEMI-ATOMISTIC MODELLING OF G4s

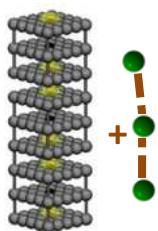
Model

- MD simulations with a Langevin thermostat
- All interactions represented via a reduced set of interaction potentials

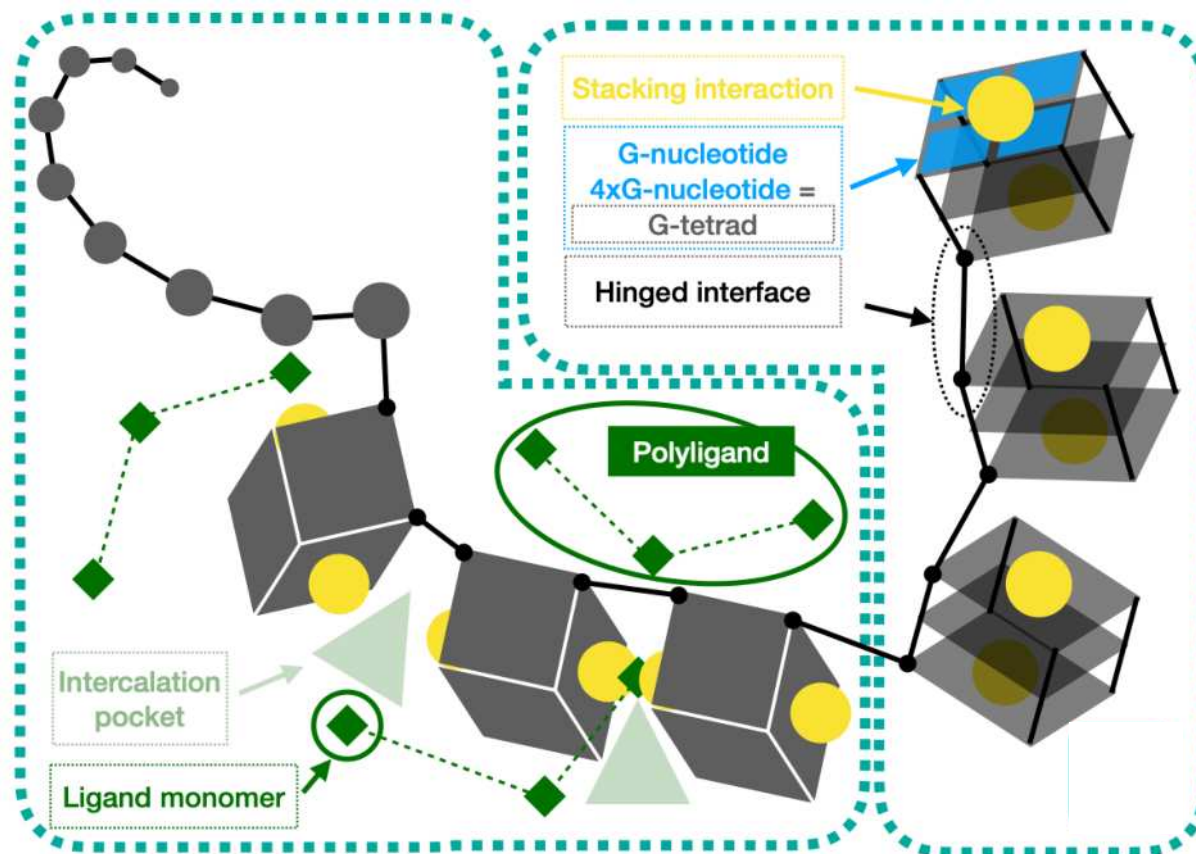


Mostarac, D. et al. Biomacromolecules (2025),
<https://doi.org/10.1021/acs.biomac.5c00176>
 arXiv:2408.07415 2024





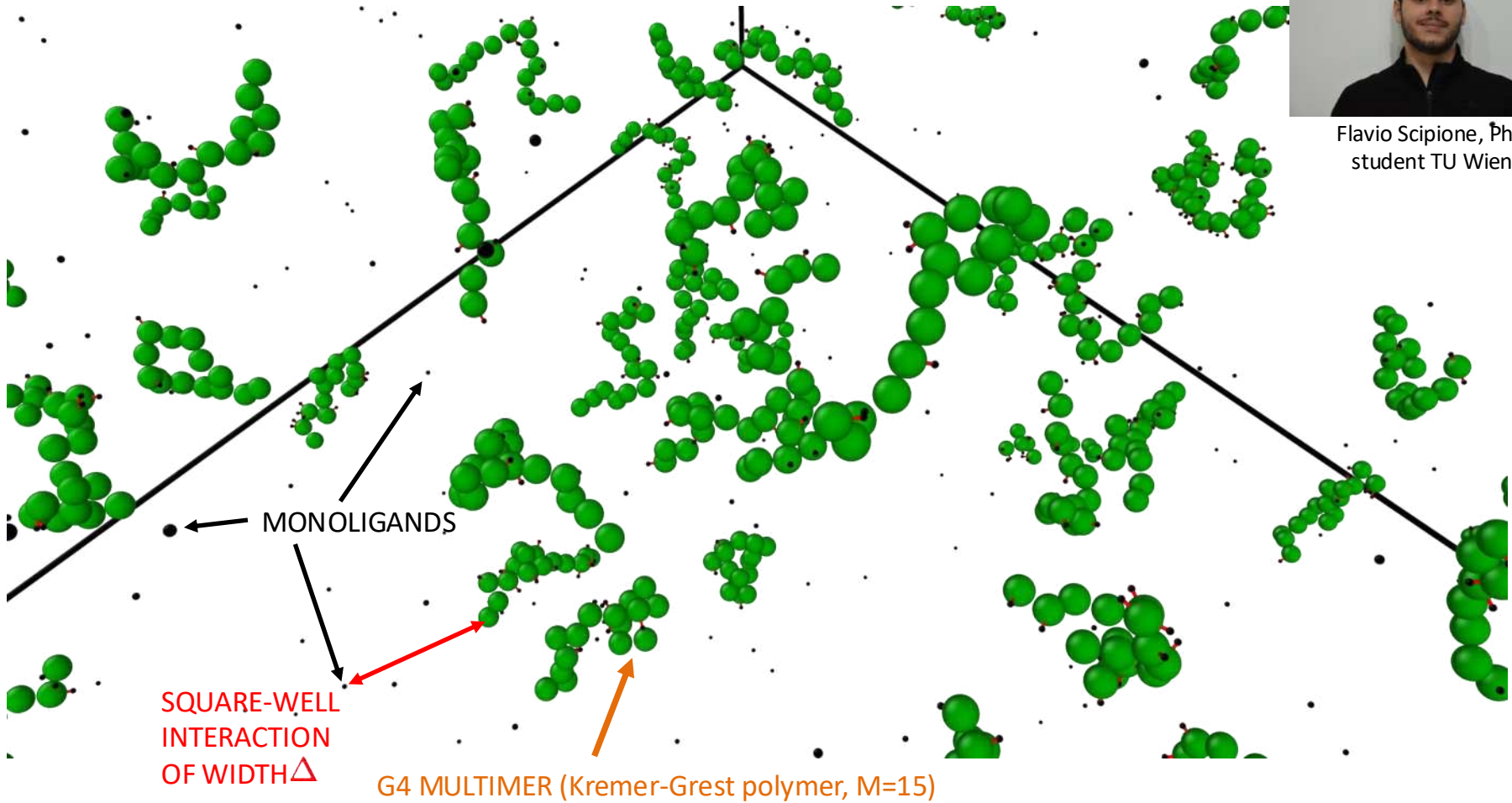
SEMI-ATOMISTIC MODEL WITH POLYLIGANDS



PUSHING COARSE-GRAINING EVEN FURTHER



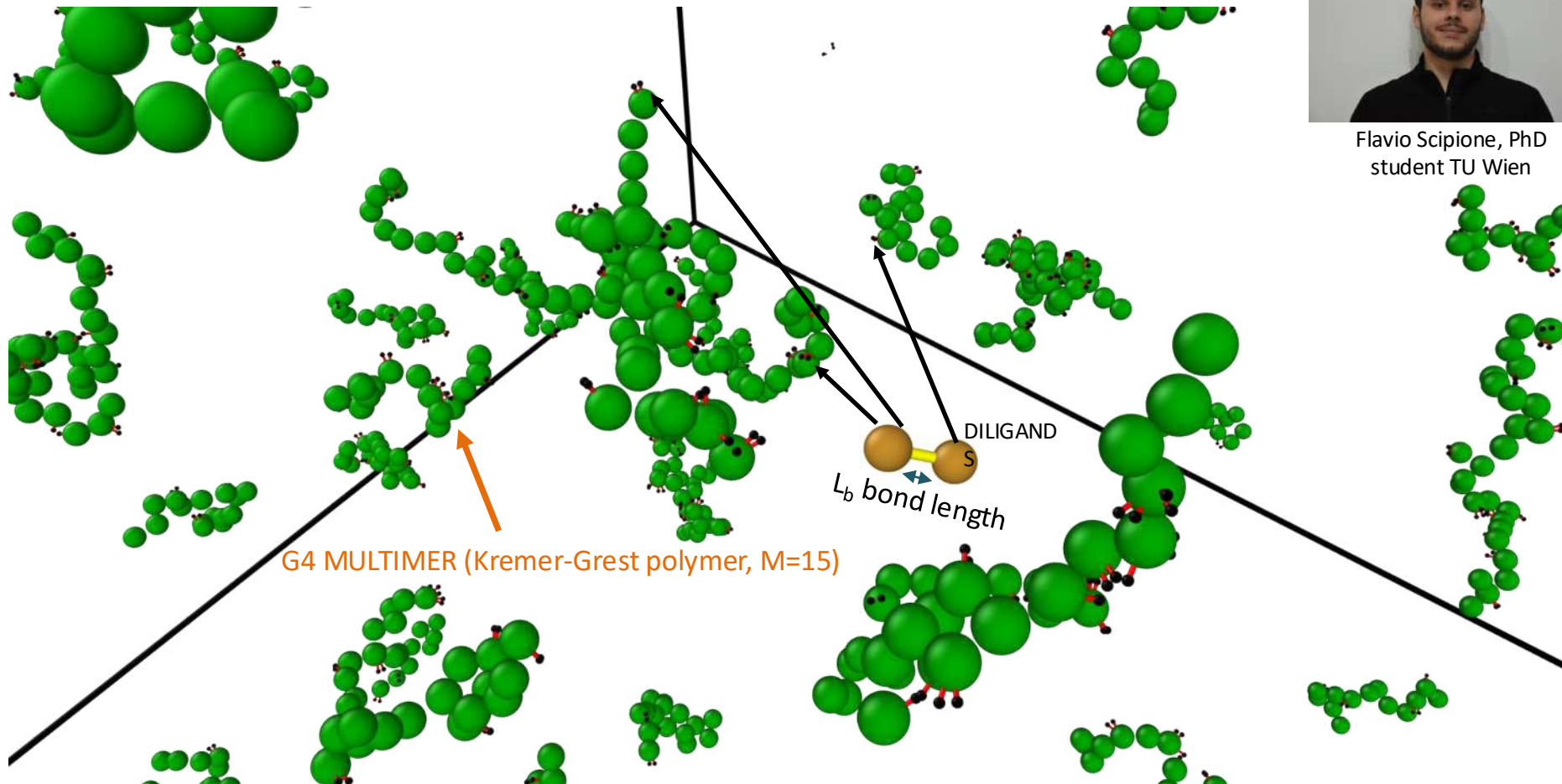
Flavio Scipione, PhD
student TU Wien



PUSHING COARSE-GRAINING EVEN FURTHER

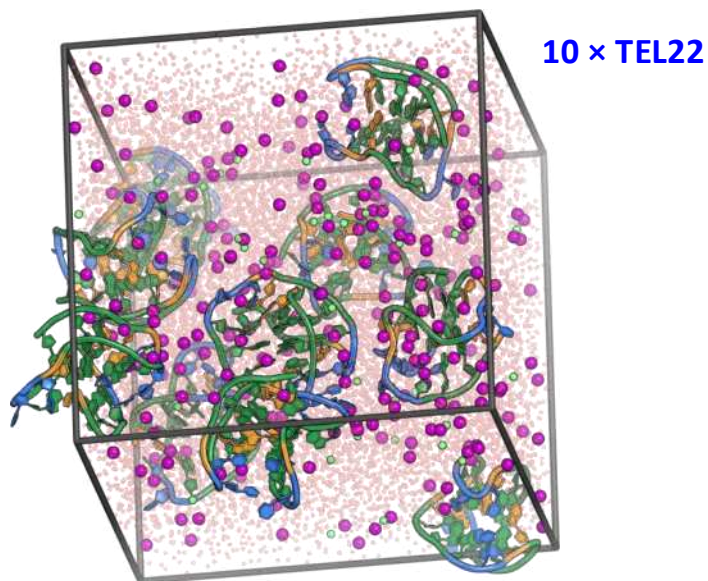


Flavio Scipione, PhD
student TU Wien



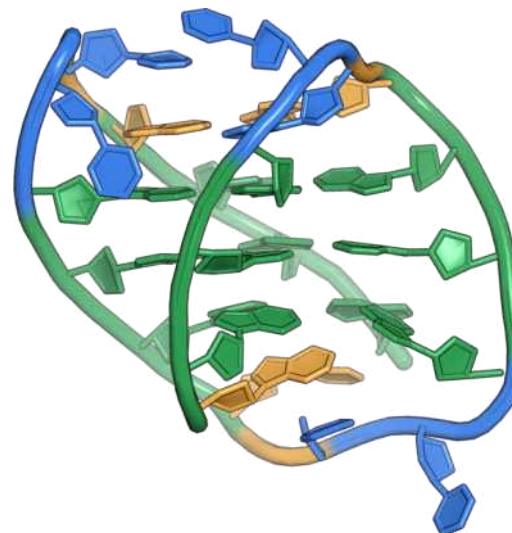
TEL22 G-QUADRUPLEX IN WATER

the all-atom reference system (GROMACS)



SIMULATION BOX: 7.9 nm, PBC

10 × TEL22 + 14 162 H₂O + 256 K⁺ + 46 Cl⁻ = 49 938 atoms



TEL22 = AGGG(TTAGGG)₃

PDB 143D, 3 G-tetrads: G A T

Force field: AMBER99SB-ILDN, TIP3P water

Ions: K⁺ to neutralize + ~0.15 M KCl

Electrostatics: PME, cut-off 1.0 nm

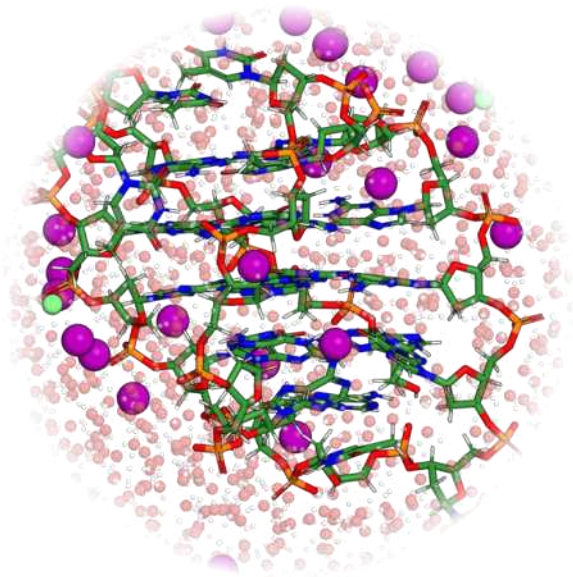
NPT: 300 K (V-rescale), 1 bar (Parrinello–Rahman)

Integration: dt = 2 fs, H-bond constraints

Every 1 ps: positions *and* forces → force-matching dataset

ML COARSE-GRAINED MODELS

learning the potential of mean force



ALL-ATOM + EXPLICIT WATER

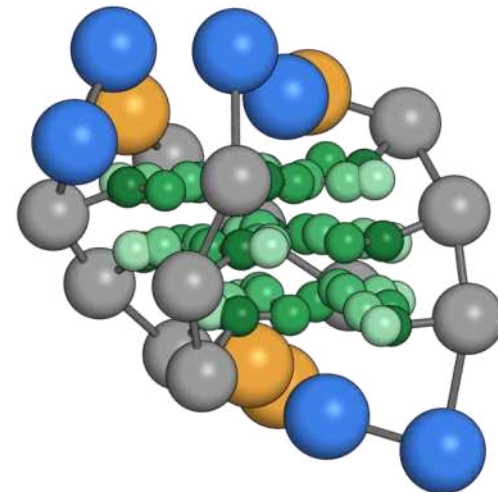
10 × TEL22 + K⁺, Cl⁻: ~50 000 atoms

~1.5 · 10⁵ degrees of freedom

$$\mathbf{R} = \mathbf{M}(\mathbf{r})$$



integrate out water, ions
and atomistic DOF



CG, IMPLICIT SOLVENT

A, T → 1 bead; G → rigid body (6 sites)

~10³ degrees of freedom

many-body PMF

$$U_{\text{CG}}(\mathbf{R}) = -k_B T \ln \int d\mathbf{r} \delta(\mathbf{M}(\mathbf{r}) - \mathbf{R}) e^{-\beta u(\mathbf{r})}$$

force matching

$$\chi^2[\theta] = \left\langle \left| \mathbf{M}_F \mathbf{f}(\mathbf{r}) + \nabla U_\theta(\mathbf{M}(\mathbf{r})) \right|^2 \right\rangle_{\text{AA}} \longrightarrow -\nabla U_\theta \simeq -\nabla U_{\text{CG}} \text{ (mean force)}$$

PaiNN-LT model

$$U_\theta(\mathbf{R}) = U_{\text{prior}}(\mathbf{R}) + U_{\text{PaiNN}}(\mathbf{R}; \theta) \quad \text{priors (bonds, angles, WCA) + equivariant GNN}$$

Noid et al., JCP 128, 244114 (2008) · CGnet: ACS Cent. Sci. 5, 755 (2019) · CGSchNet: JCP 153, 194101 (2020) · PaiNN: ICML (2021)

EXAMPLES OF COLLOIDAL PARTICLES

tetrapods

Dimeri (sim./asim.)

fused clusters

cubes

triangular prisms

rods/disks

ellipsoids

Janus particles

Patchy spheres

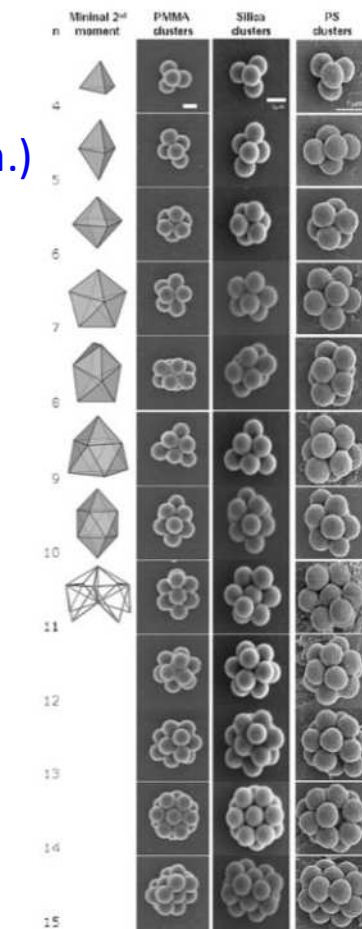


Figure 2. Minimal moment models and scanning electron micrographs of polystyrene, silica, and PMMA colloidal clusters. Scale bars: micrometer.

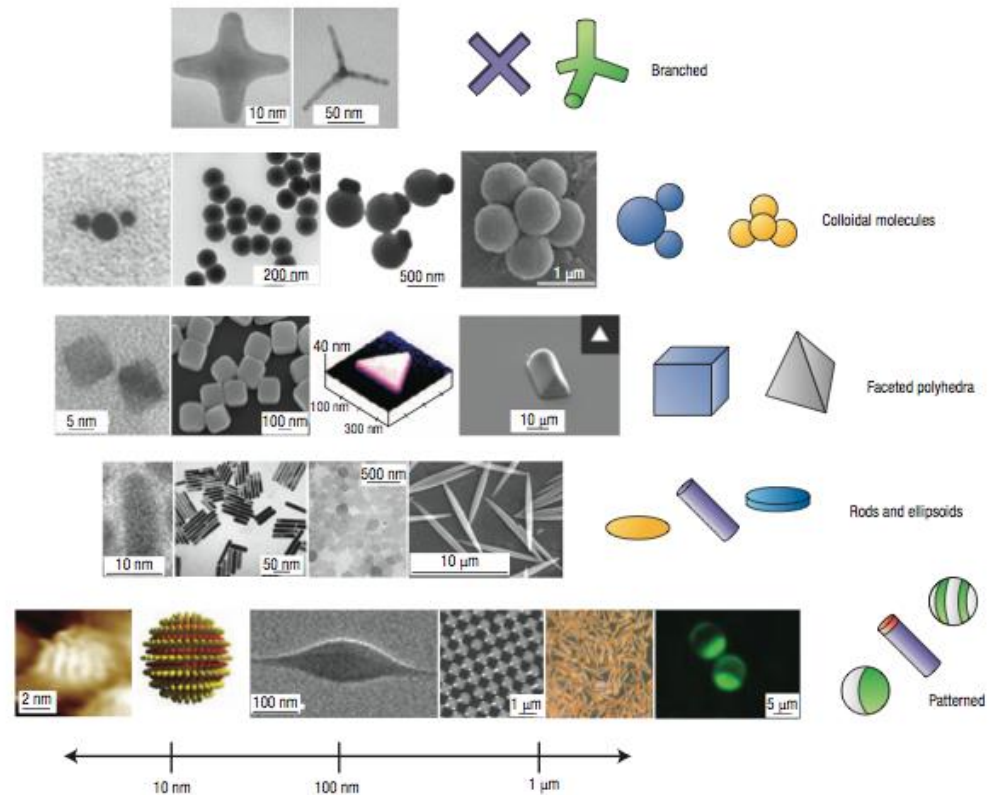
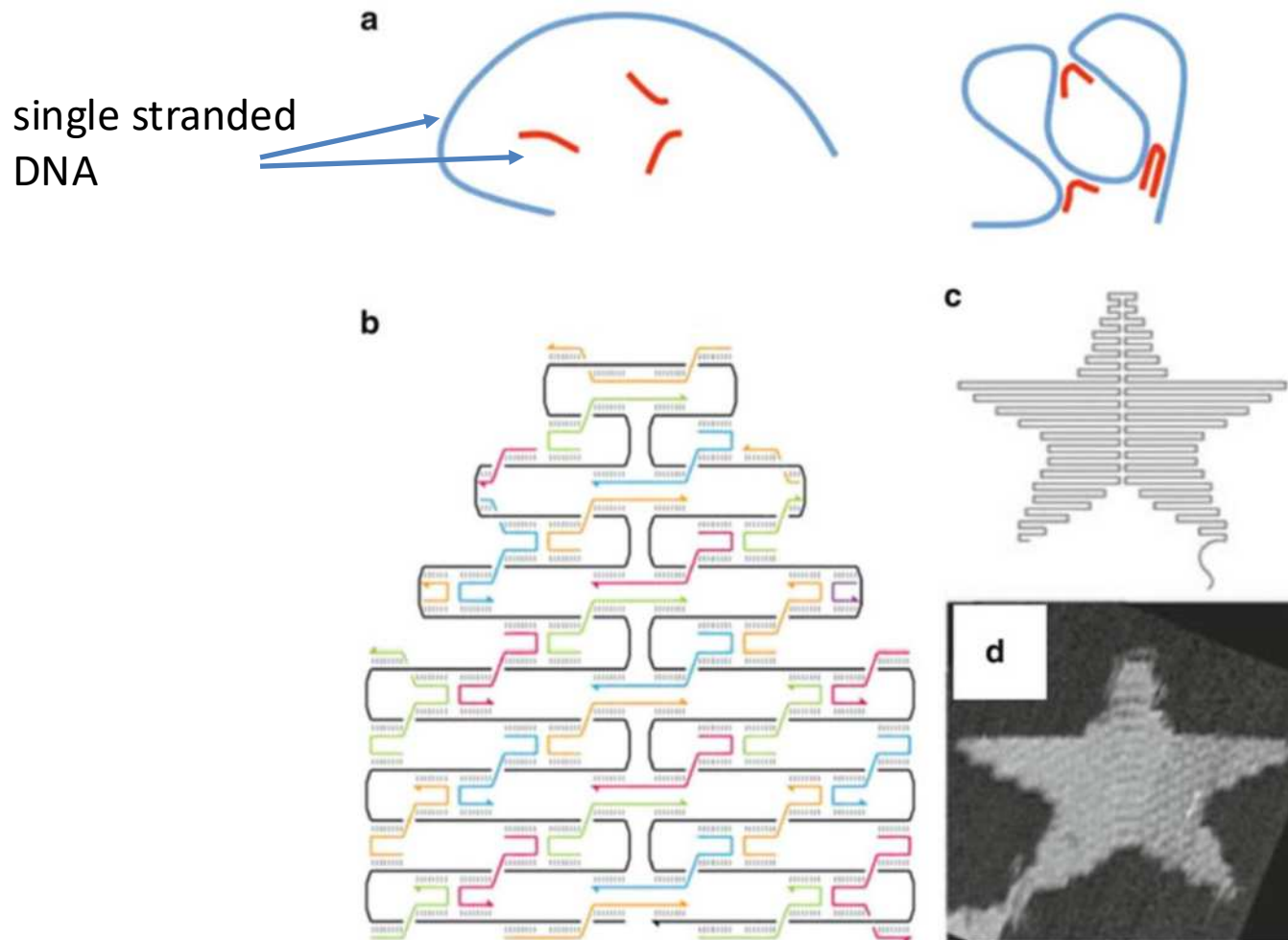


Figure 1 Representative examples of recently synthesized anisotropic particle building blocks. The particles are classified in rows by anisotropy type and increase in size from left to right according to the approximate scale at the bottom. From left to right, top to bottom: branched particles include gold⁵¹ and CdTe⁷¹ tetrapods. DNA-linked gold nanocrystals⁵² (the small and large nanocrystals are 5 nm and 10 nm respectively), silica dumb-bells⁷², asymmetric dimers⁷³ and fused clusters⁷⁷ form colloidal molecules. PbSe⁷⁴ and silver cubes¹⁰ as well as gold⁷⁶ and polymer triangular prisms⁵³ are examples of faceted particles. Rods and ellipsoids of composition CdSe⁷⁵, gold⁷⁸, gibbsite⁴ and polymer latex⁶⁰ are shown. Examples of patterned particles include striped spheres⁷⁷, biphase rods¹⁴, patchy spheres with 'valence'³⁴, Au–Pt nanorods⁷⁸ (the rod diameters are of the order of 200–300 nm) and Janus spheres⁷⁹. Images reprinted with permission from the references as indicated. Copyright, as appropriate, AAAS, ACS, RSC, Wiley-VCH.

G. Yi *et al.* Adv. Mater. **16**, 1204 (2004).

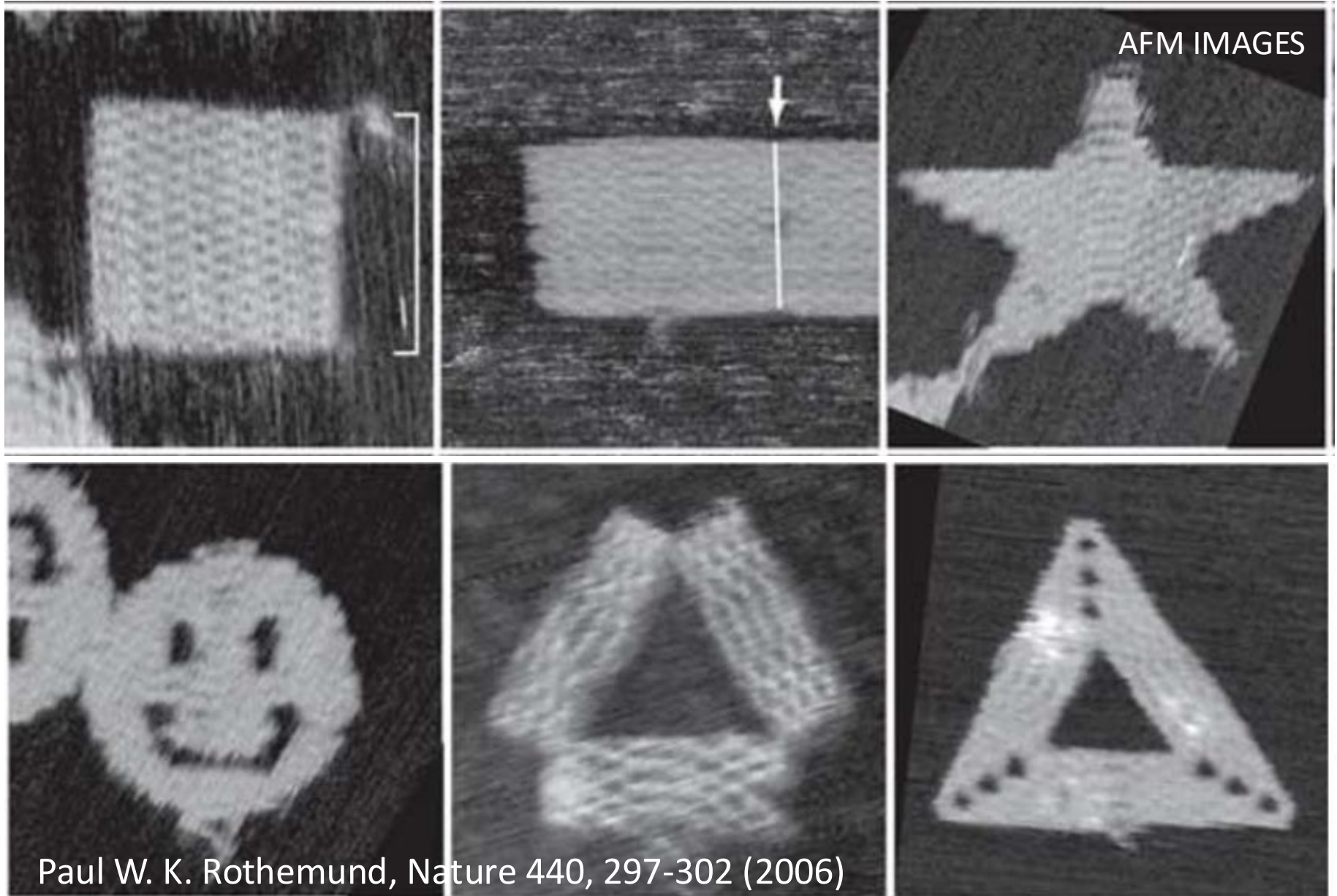
S. C. Glotzer and M. J. Solomon, Nat. Mat. **6**, 557 (2007).

CREATING MOLECULES WITH DNA: DNA ORIGAMI



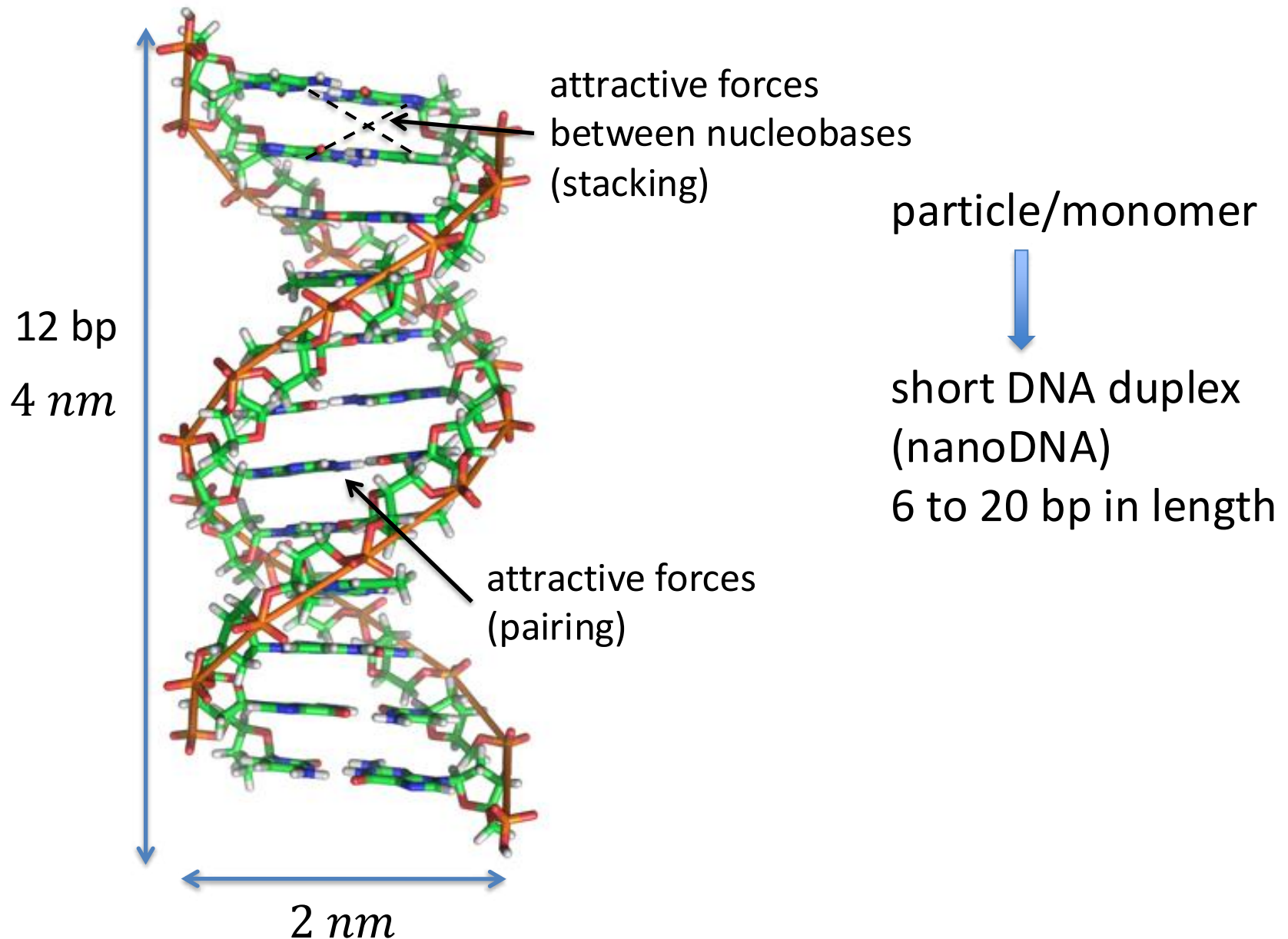
a) Sketch of the basic idea of DNA origami: a long strand is “stapled” by oligonucleotides complementary to specific tracts along its sequence and is folded into designed shapes (b), as complex as a star (c). (d) AFM image of the experimental realization of (c) (edge size is 165 μm).

EXAMPLES OF DNA ORIGAMI



Paul W. K. Rothemund, Nature 440, 297-302 (2006)

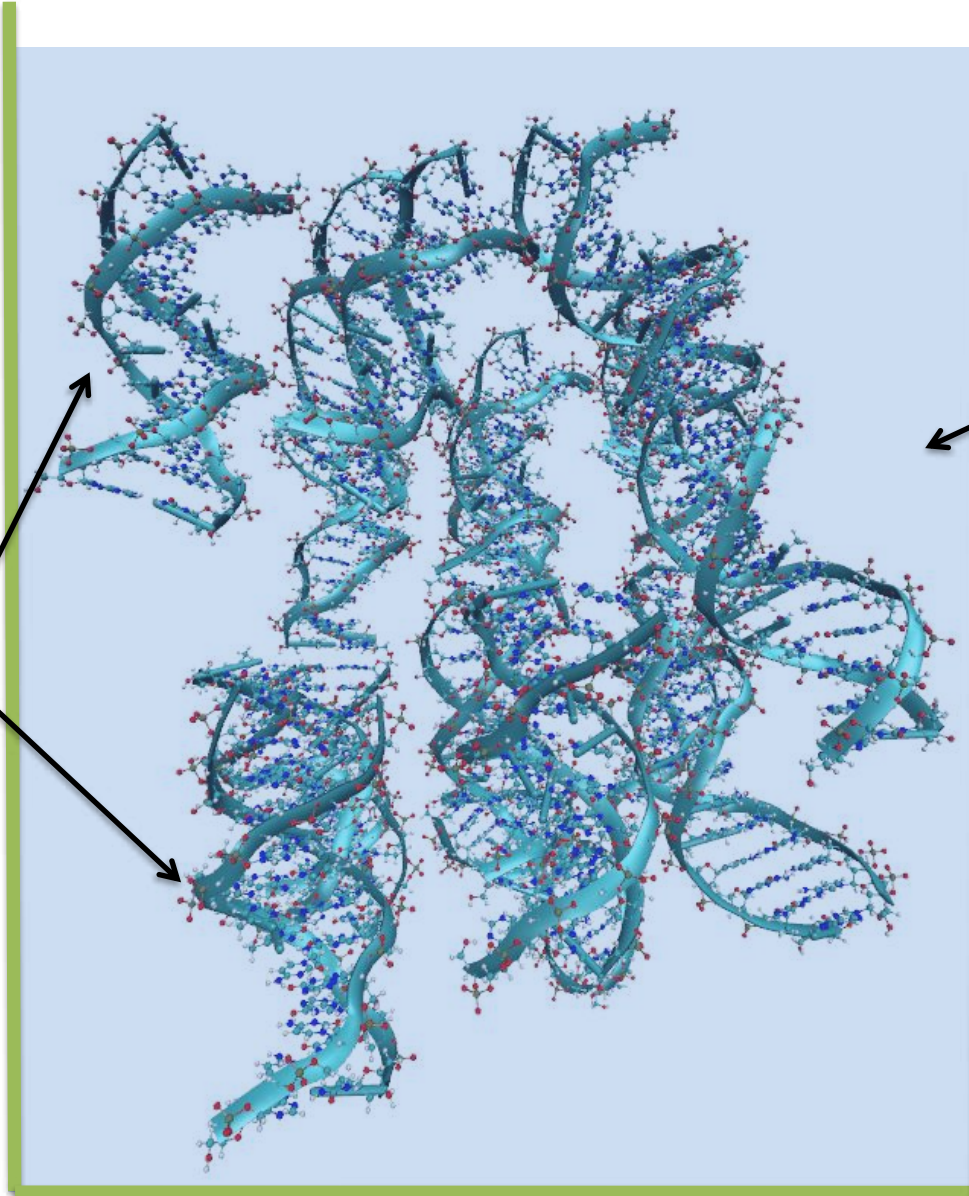
DNA NANODUPLEXES



EXPERIMENTAL SYSTEM

DNA DOUBLE
HELICES

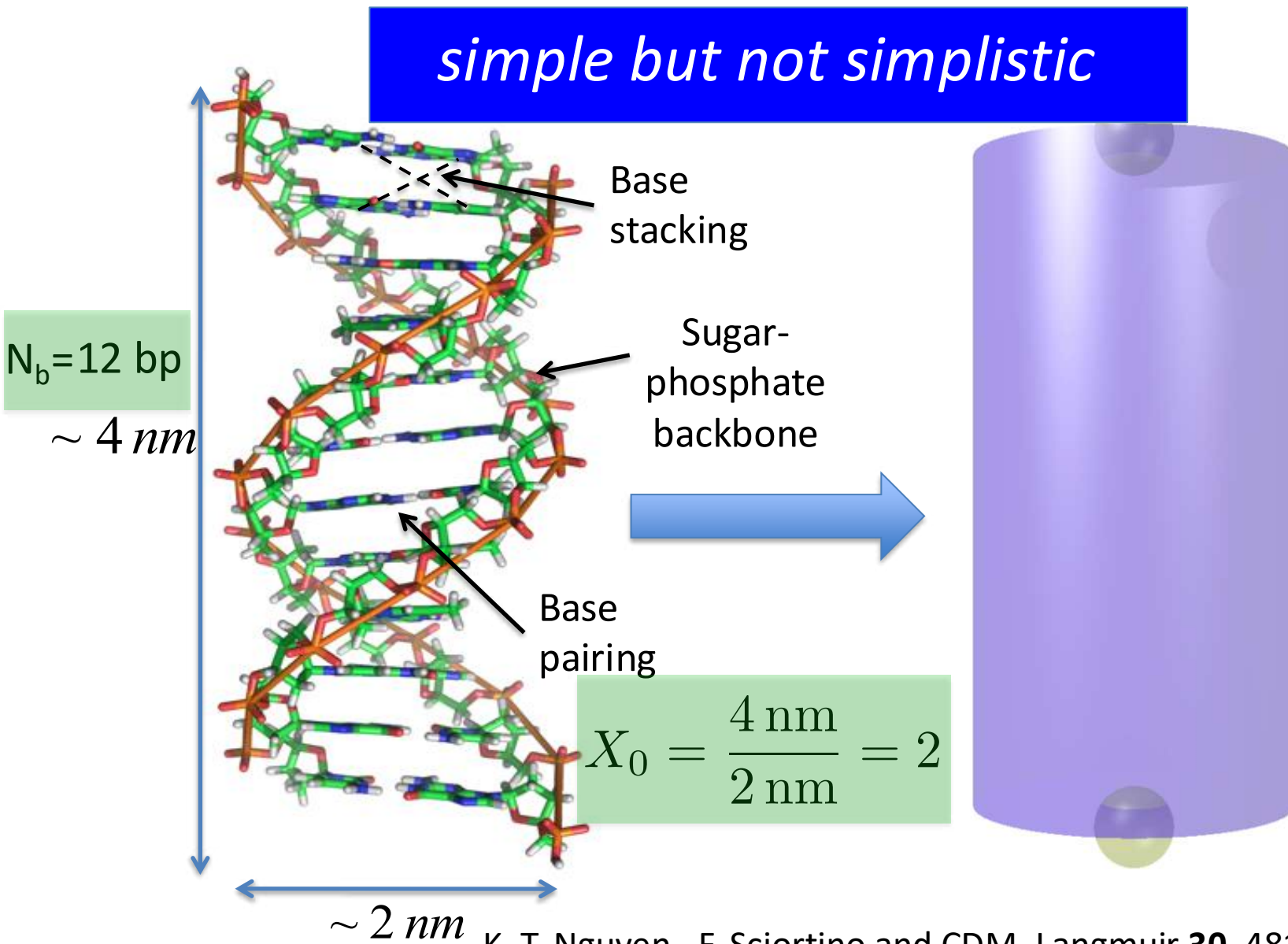
WATER



M. Nakata *et al.* Science **318**, 1276 (2007)

HARD CYLINDERS DNA MODEL

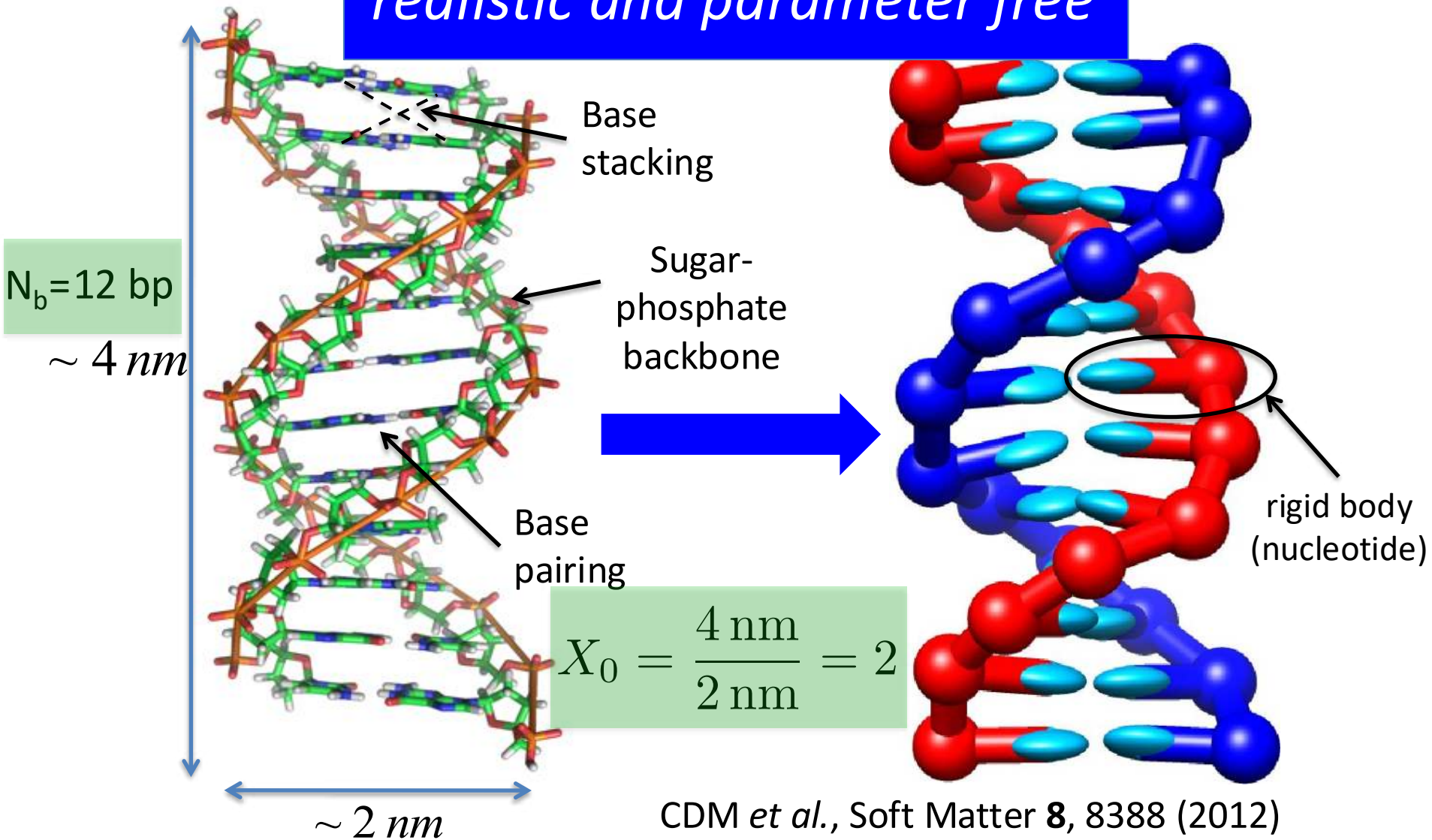
simple but not simplistic



K. T. Nguyen, F. Sciortino and CDM, Langmuir **30**, 4814 (2014)

OXFORD DNA MODEL

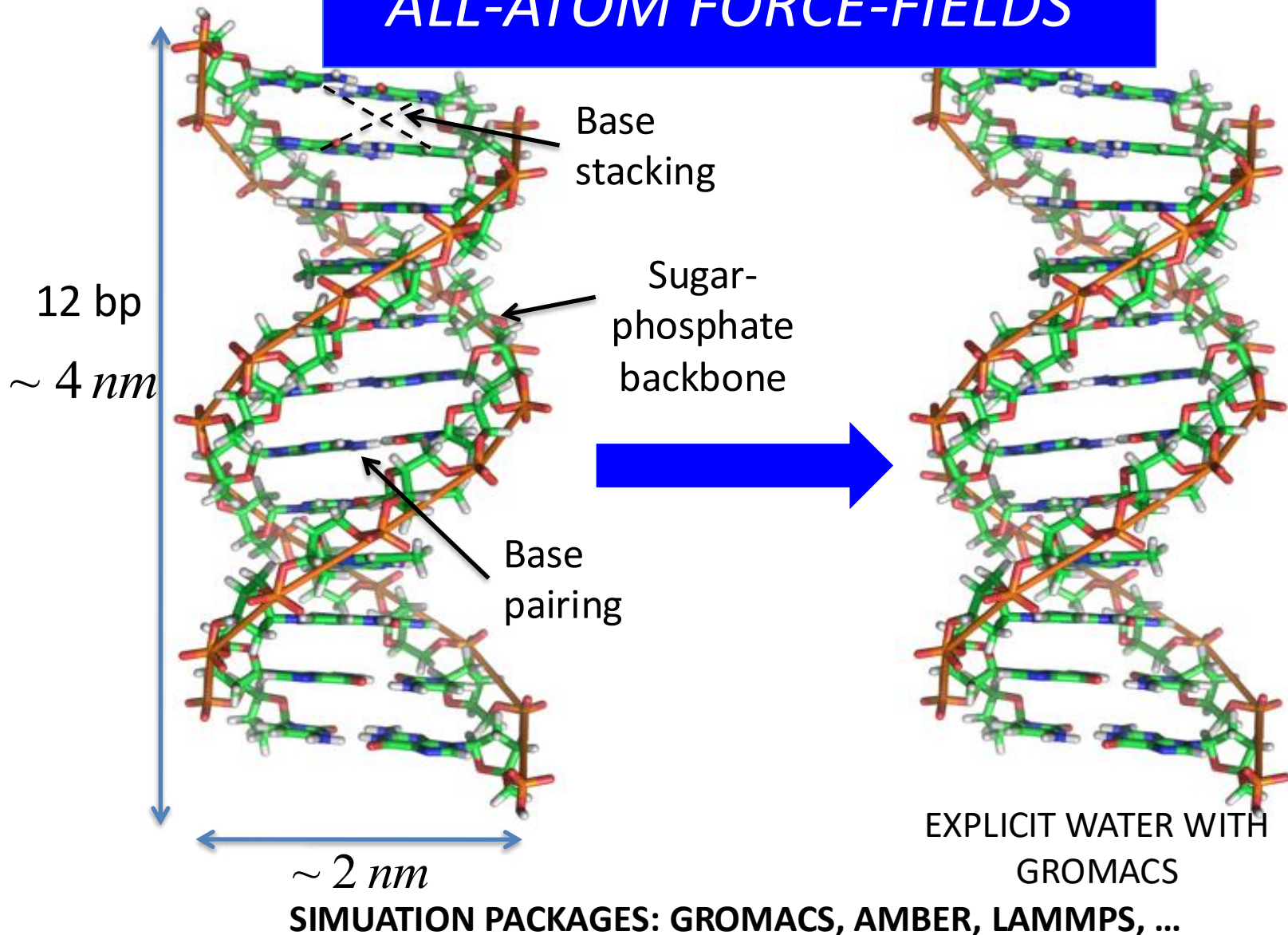
realistic and parameter free



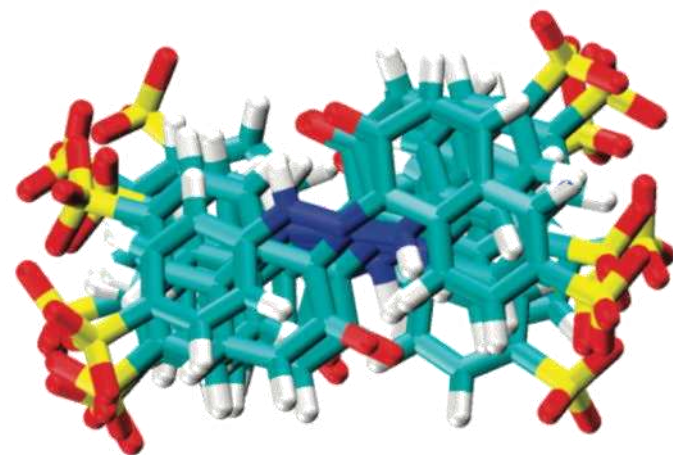
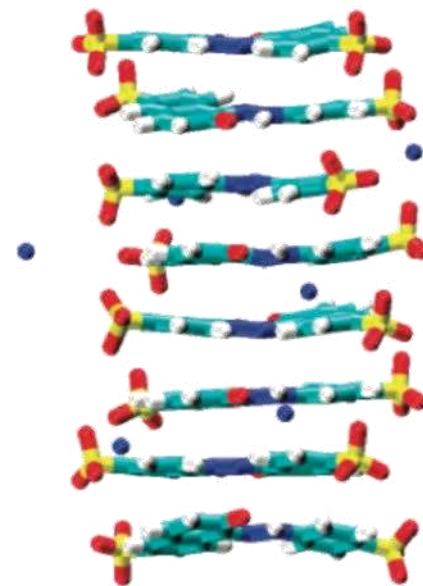
CDM *et al.*, Soft Matter **8**, 8388 (2012)

DNA FULL-ATOM MODEL

ALL-ATOM FORCE-FIELDS

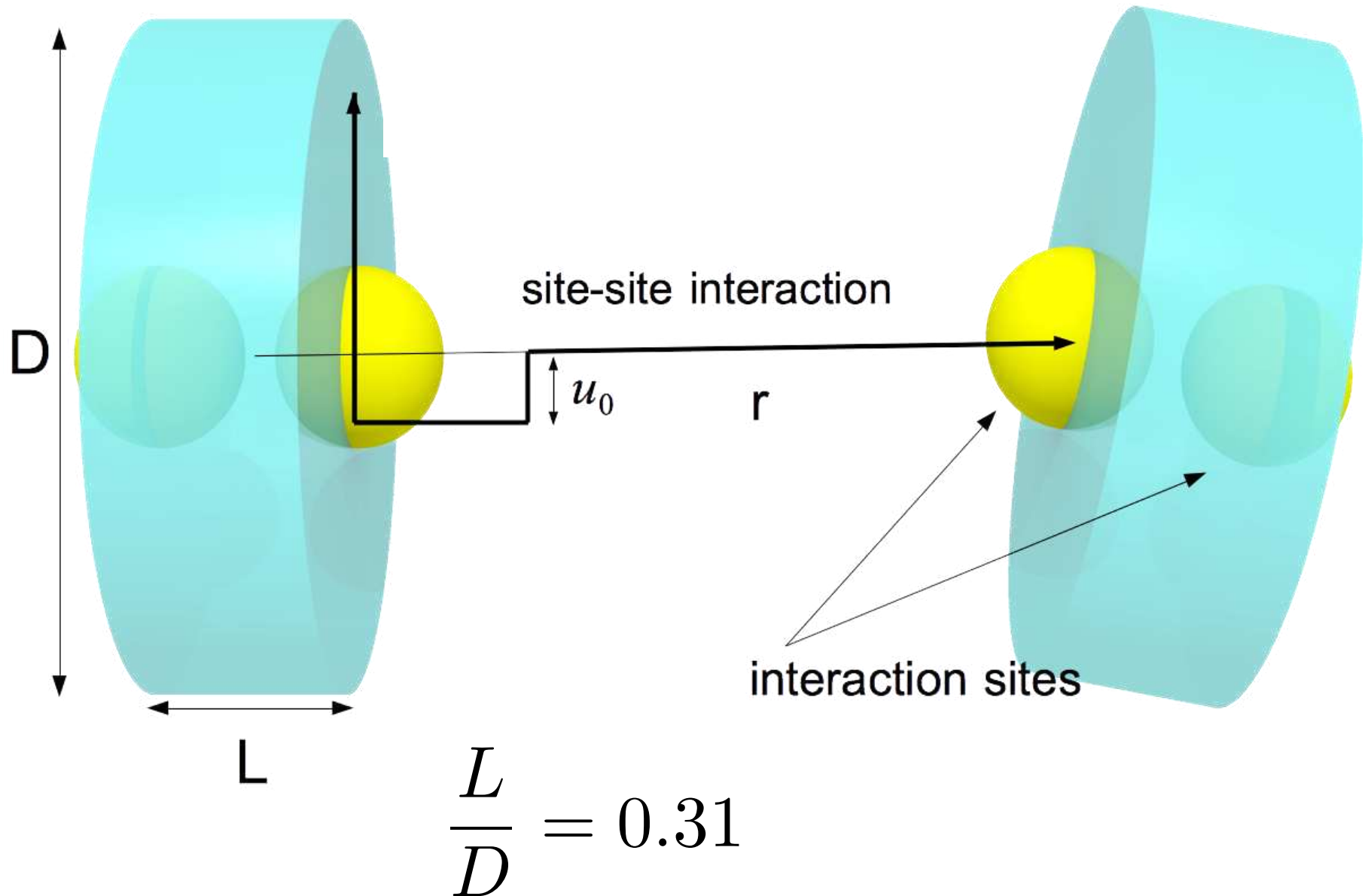


CHROMONICS: SUNSET YELLOW (SSY)



F. Chami & M. R. Wilson, JACS **132**, 7794 (2010)

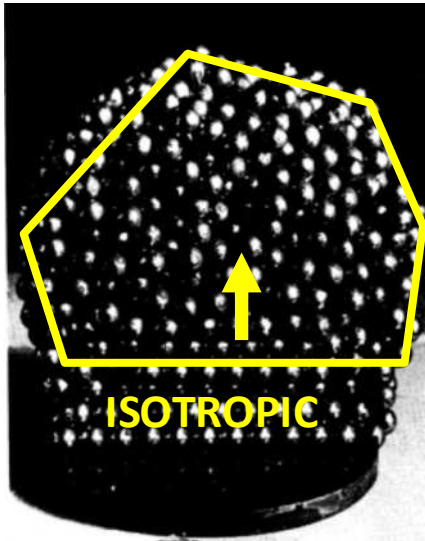
COARSE-GRAINED MODEL OF SSY



LIQUID CRYSTALS (MESOPHASES)

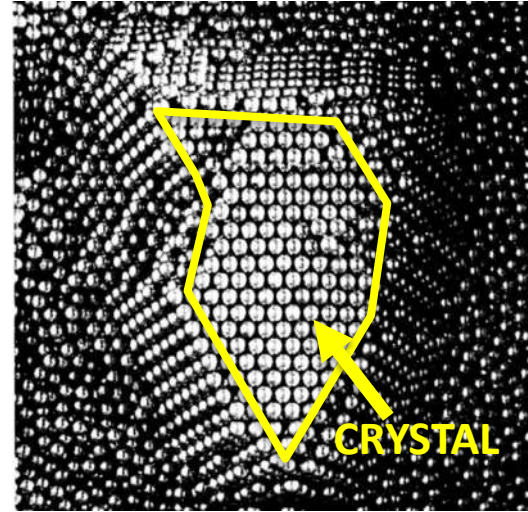


J.D. Bernal



The Bakerian Lecture, 1962

ISOTROPIC



The Bakerian Lecture, 1962

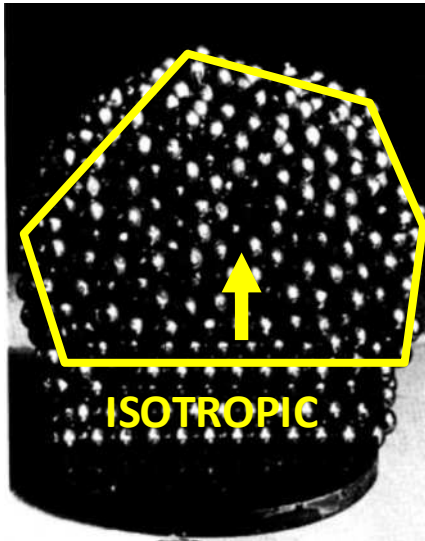
CRYSTAL

SPHERICAL PARTICLES

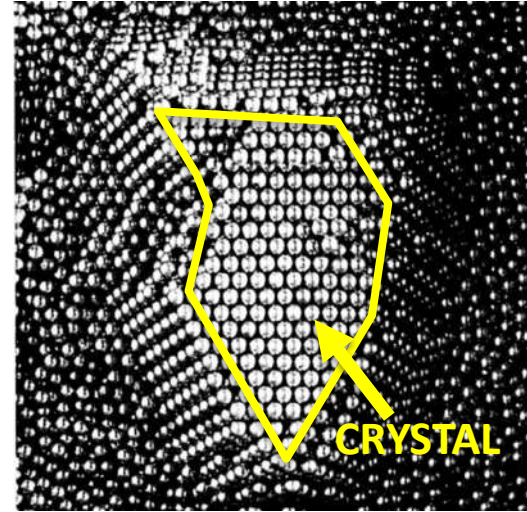
LIQUID CRYSTALS (MESOPHASES)



J.D. Bernal



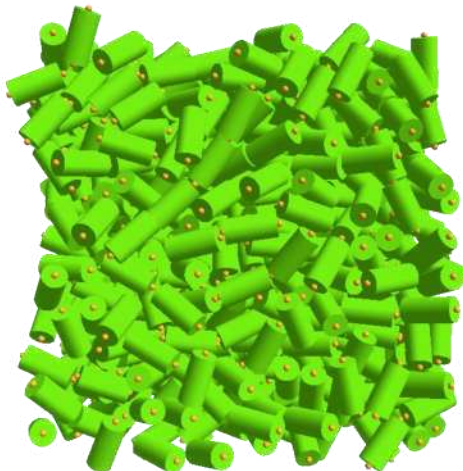
The Bakerian Lecture, 1962



The Bakerian Lecture, 1962

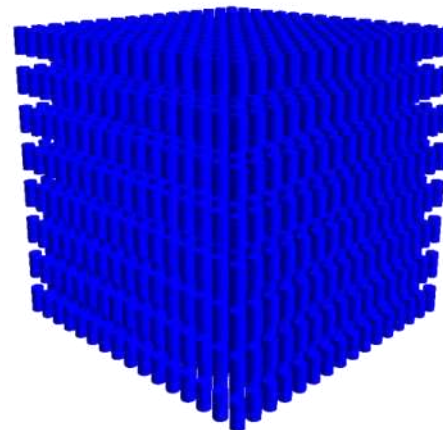
SPHERICAL PARTICLES

ISOTROPIC



ISOTROPIC

CRYSTAL



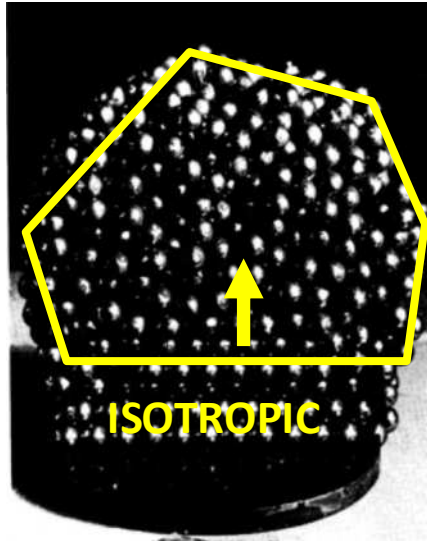
CRYSTAL

ELONGATED PARTICLES

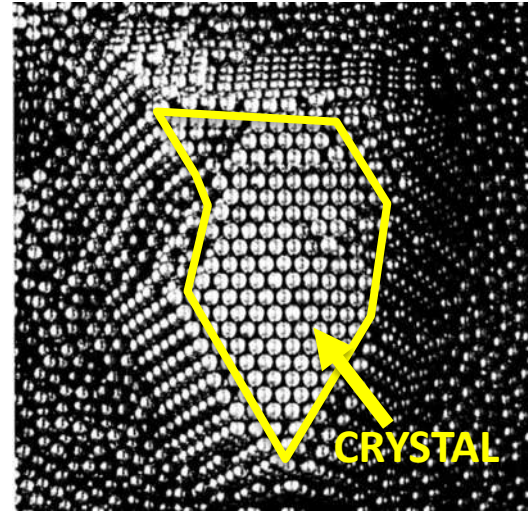
LIQUID CRYSTALS (MESOPHASES)



J.D. Bernal



The Bakerian Lecture, 1962

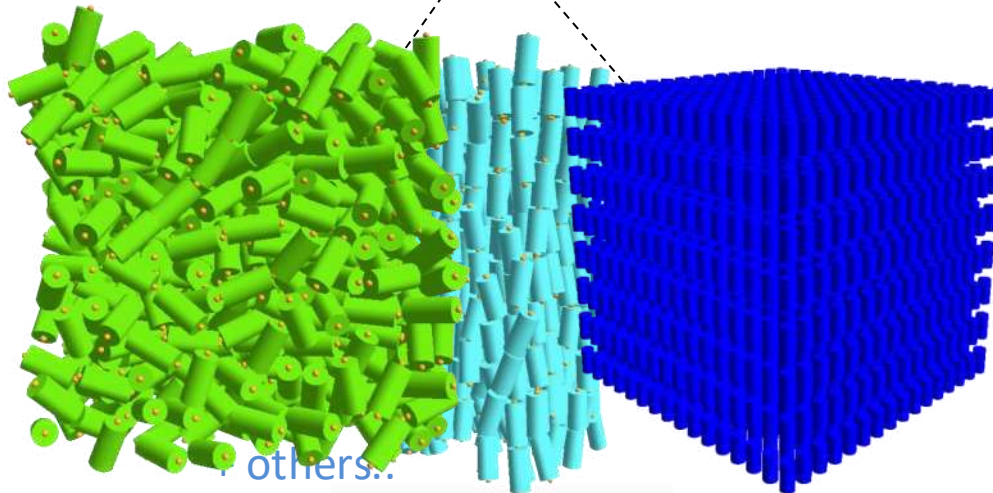


The Bakerian Lecture, 1962

SPHERICAL PARTICLES

ISOTROPIC

CRYSTAL

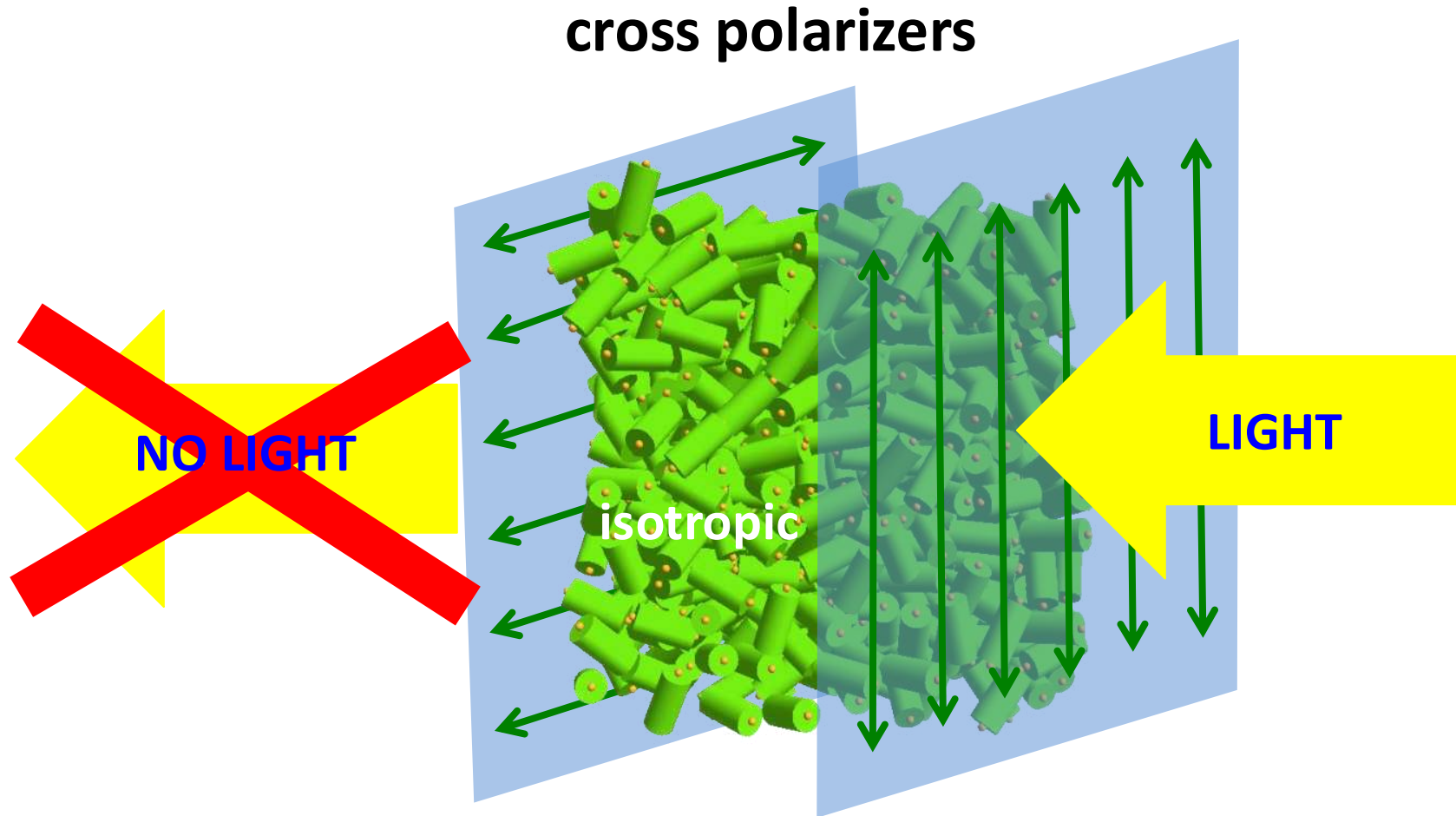


ISOTROPIC

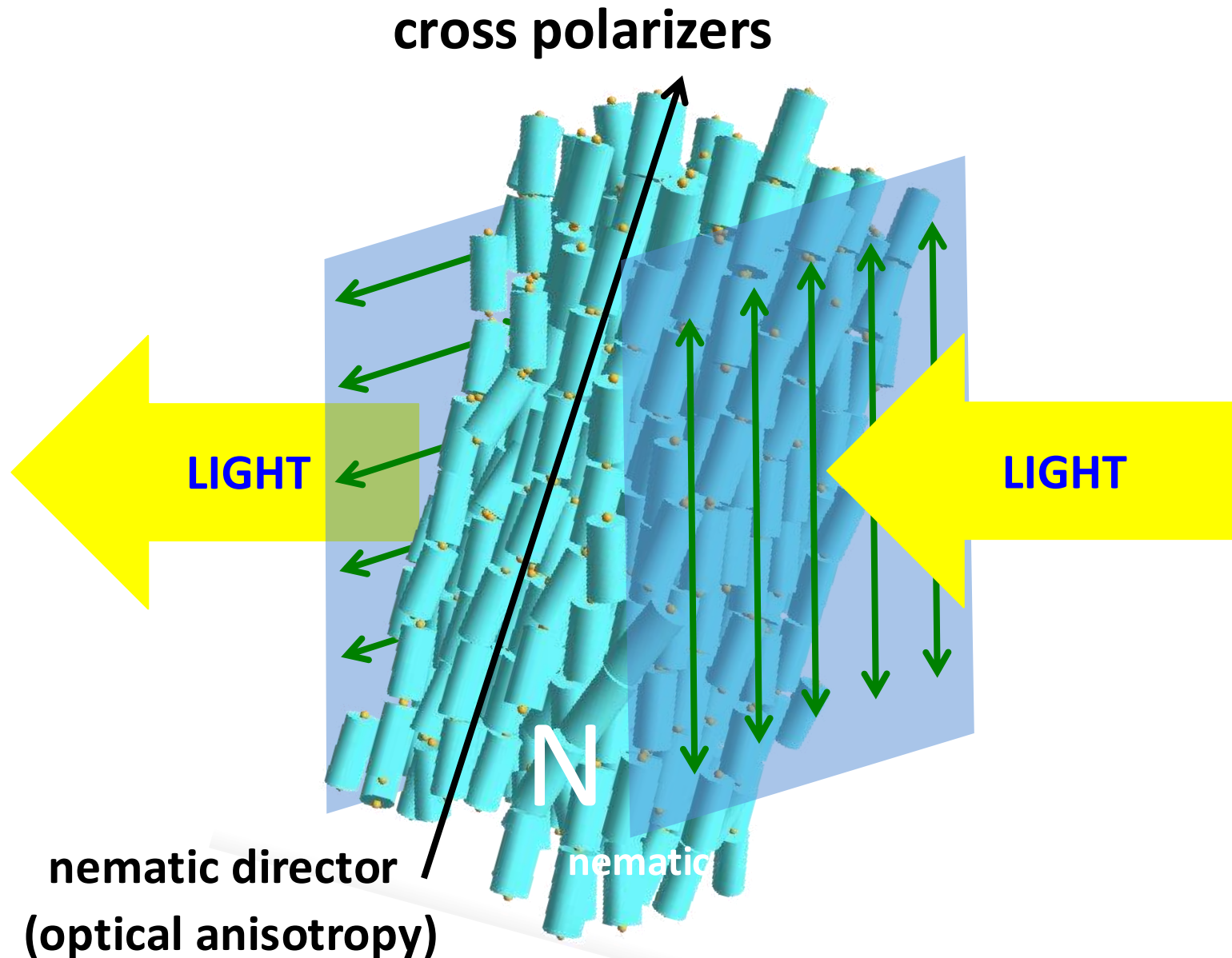
NEMATIC MESOPHASE CRYSTAL

ELONGATED PARTICLES

BIREFRINGENCE OF LIQUID CRYSTALS

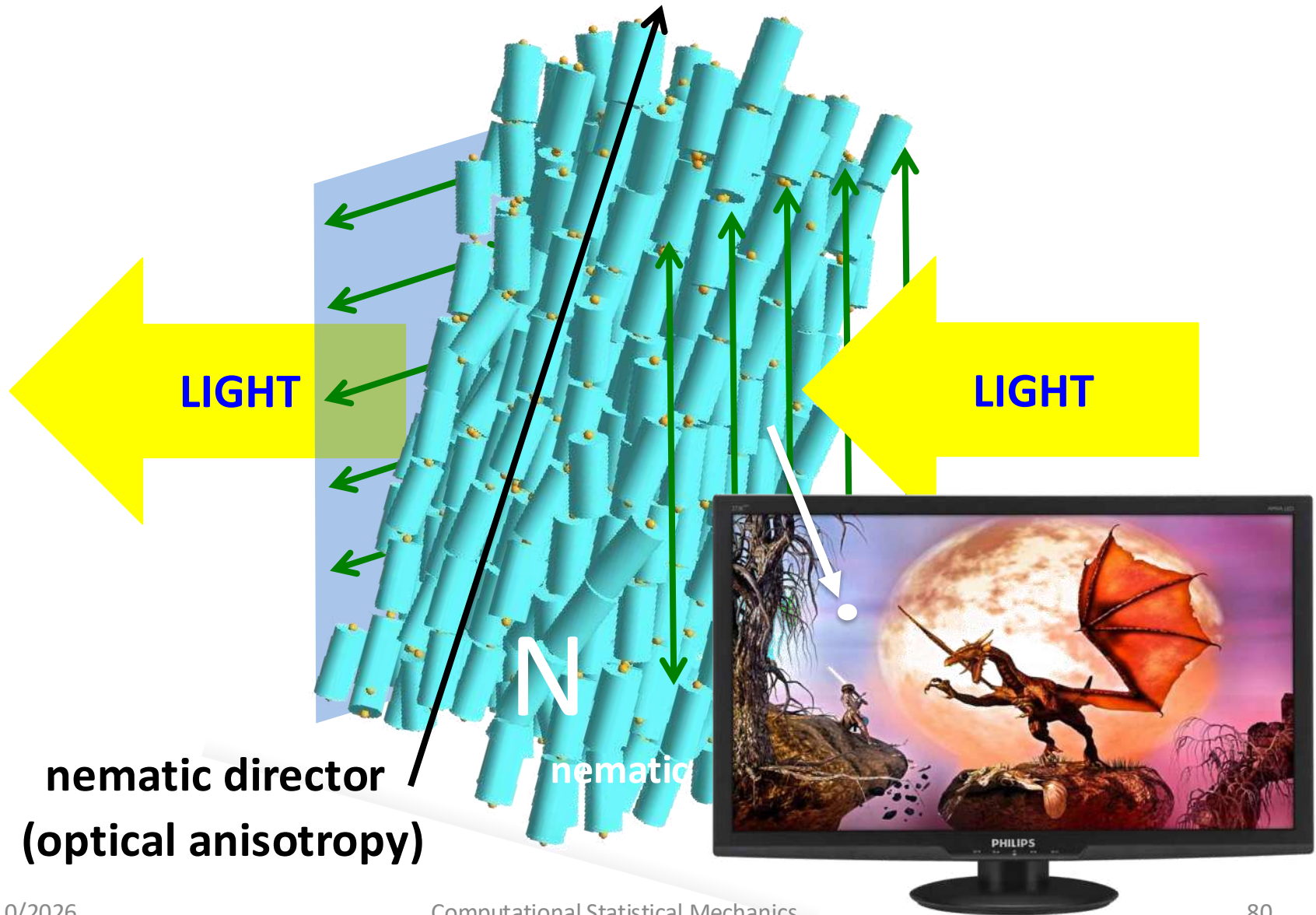


BIREFRINGENCE OF LIQUID CRYSTALS



BIREFRINGENCE OF LIQUID CRYSTALS

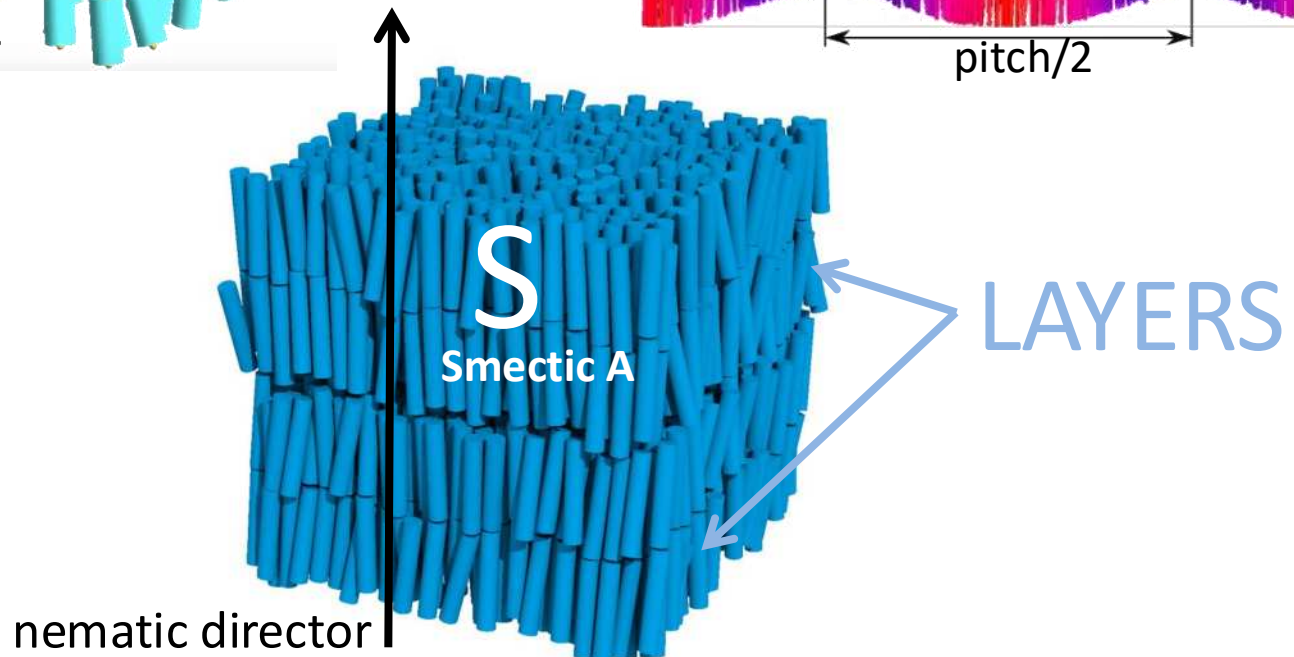
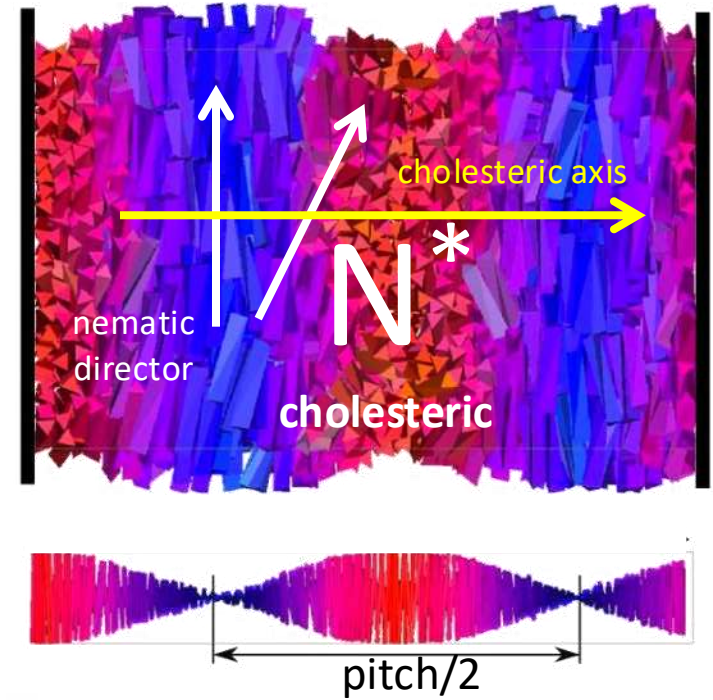
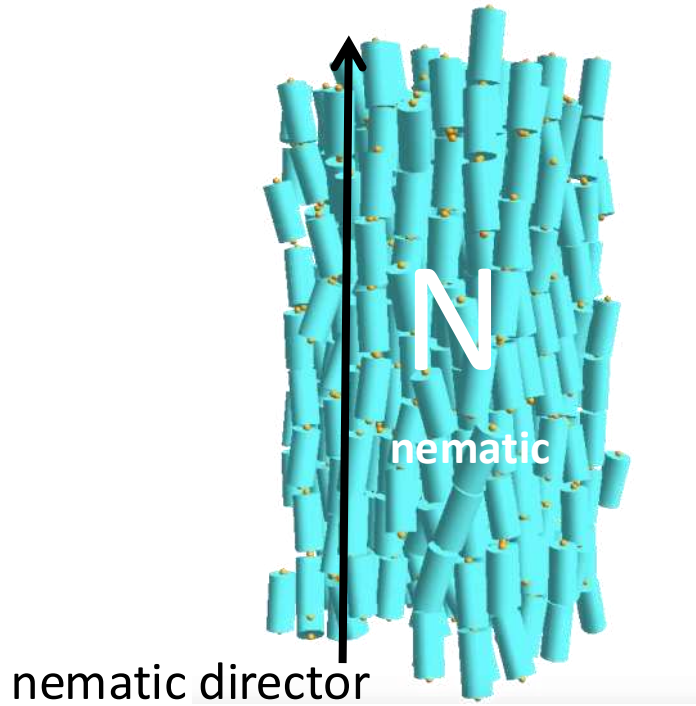
cross polarizers



nematic director
(optical anisotropy)

nematic

NEMATIC, CHOLESTERIC AND SMECTIC PHASES



SMECTIC PHASES OF DNA DUPLEXES

ISOTROPIC



fusilli

NEMATIC

nematic axis

long holed fusilli of Gragnano

CHOLESTERIC

Helical axis

nematic axis

long holed fusilli of Gragnano

SMECTIC

nematic axis

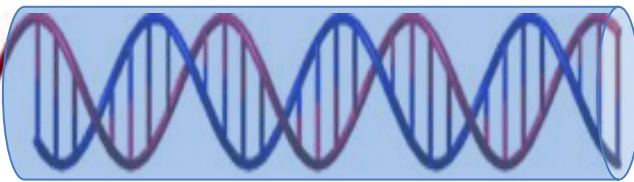
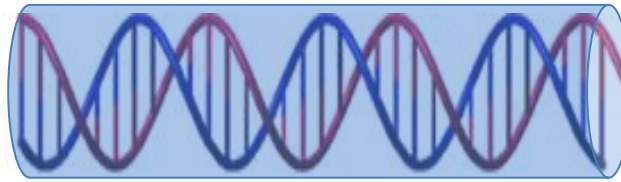
layers

fusilli

DNA NANO-NUNCHAKU

16 nm

16 nm



2 nm

STICK

CHAIN

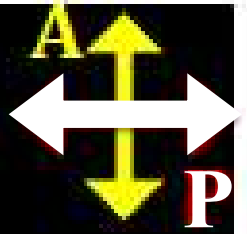
STICK

CHAIN

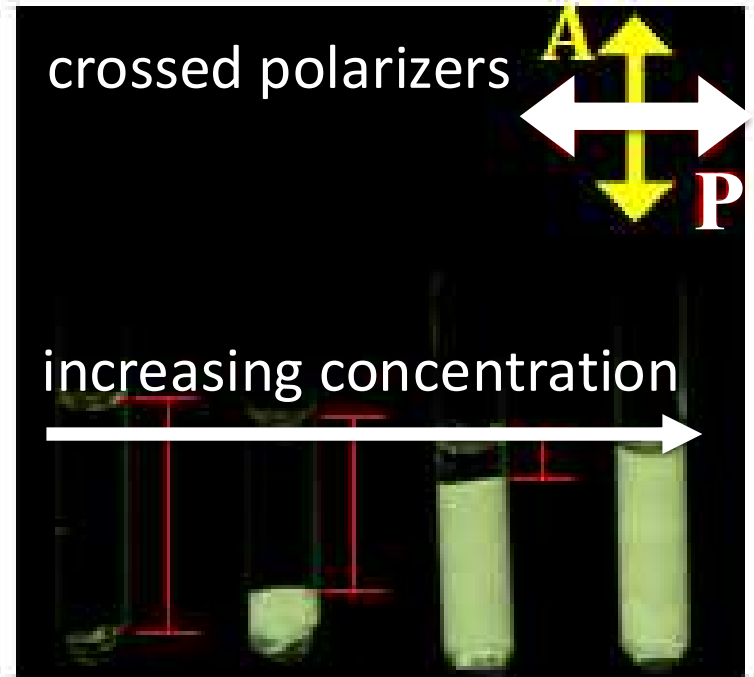


STICK

crossed polarizers

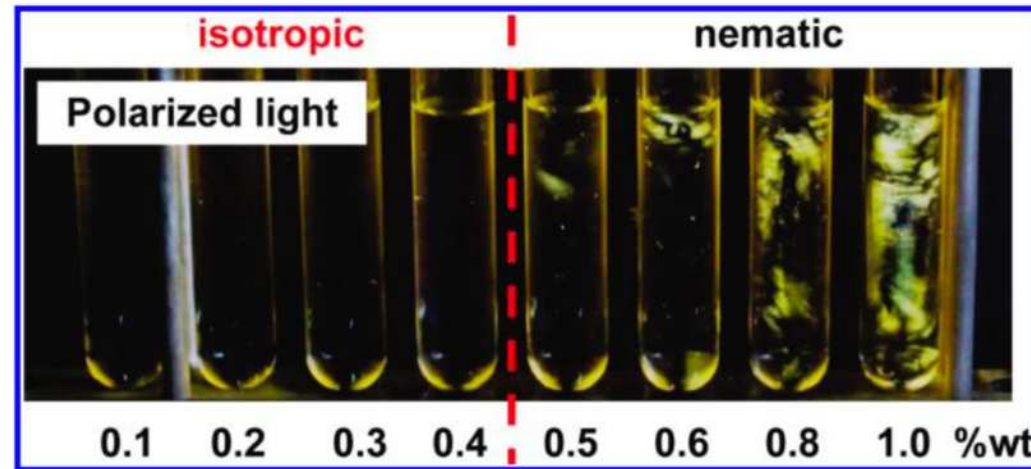
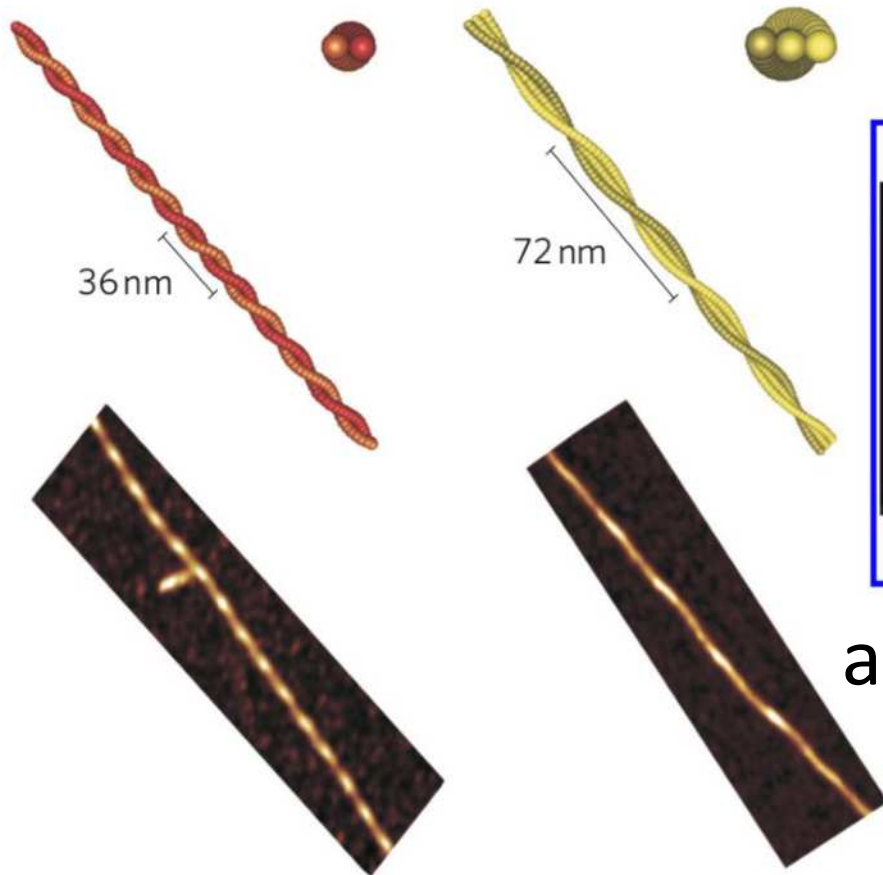


increasing concentration



NEMATIC PHASE OF LONG AMYLOID FIBRILS

B-lactoglobulin

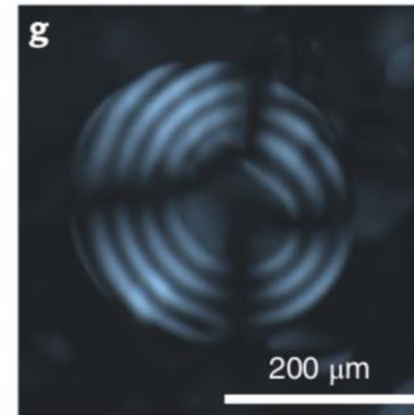
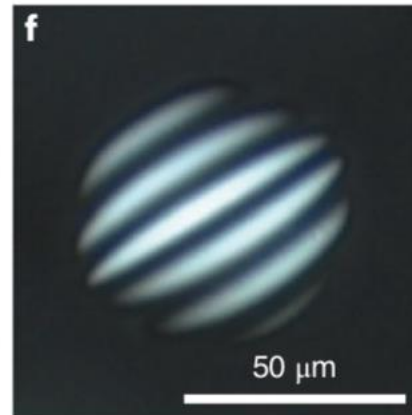
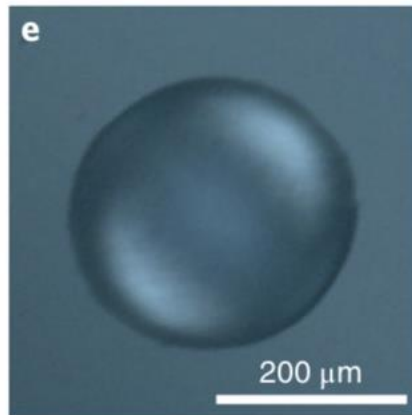
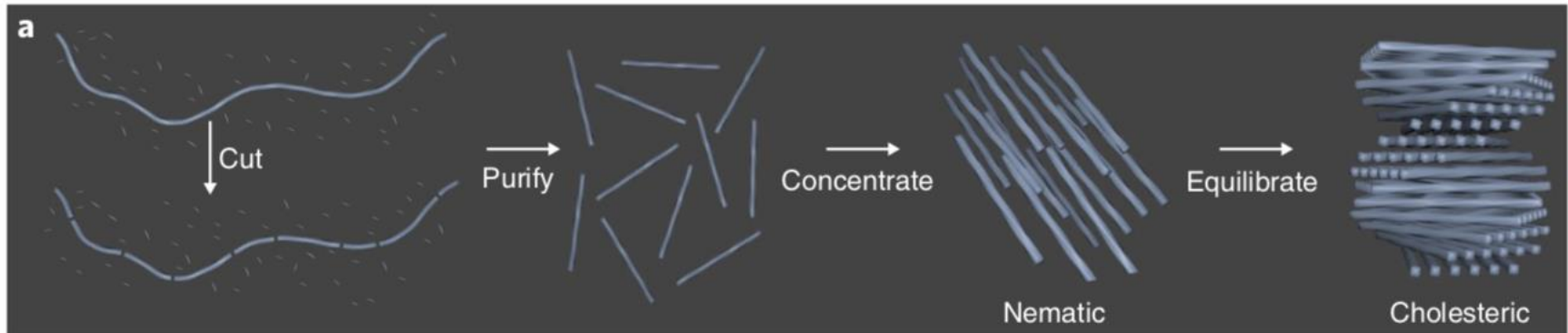


average fibril length $\approx 1\text{-}20\ \mu\text{m}$

J. Adamick *et al.* Nat. Nano. **5**, 423 (2010) R. Mezzenga *et al.* Langmuir **26** (13), 10401 (2010)

SHORT AMYLOID FIBRILS

B-lactoglobulin



average fibril length = 401 nm